

GIS-Based Industrial Development Readiness Model (IDRM)

A Spatial Decision Support Framework
for Evaluating Industrial Cities



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Executive Summary

This project develops a GIS-based Industrial Development Readiness Model (IDRM) for comparatively assessing the First Industrial City (D1), Second Industrial City (D2), and Third Industrial City (D3) in Dammam, Saudi Arabia. The study integrates macro-level, micro-level, and comparative GIS analyses covering industrial density, logistics accessibility, spatio-temporal development, Kernel Density Estimation (KDE), Standard Deviation Ellipse (SDE), and Location Quotient (LQ).

The study uses two complementary industrial datasets with deliberately different analytical roles. The macro-level Industrial Cities database contains aggregate official MODON information, including official factory counts and industrial-city areas. These values are used to calculate city-level industrial density. Separately, the Industrial Facilities database is an author-compiled balanced research sample containing 40 factories per city (120 factories in total), used for detailed temporal, spatial, and sectoral analyses including industrial growth and LQ-based specialization. The two datasets are therefore not interchangeable and are never combined for the same calculation.

The IDRM integrates five standardized indicators: Industrial Density Index (IDI), Logistics Accessibility Index (LAI), Industrial Expansion Potential Index (IEPI), Industrial Growth Index (IGI), and Industrial Specialization Index (ISI). Each indicator is standardized to a 1-5 scale and assigned an equal weight of 0.20. The resulting composite scores are D1 = 2.96, D2 = 1.93, and D3 = 3.40, producing the ranking D3 first, D1 second, and D2 third.

The central interpretation is that current industrial strength is not equivalent to future development readiness. D1 has the strongest current industrial position because of its high official factory density and superior logistics accessibility. D3 ranks highest under the proposed IDRM framework because of its substantial expansion potential, strongest absolute industrial growth within the research sample, and strong industrial specialization. The model is therefore a comparative decision-support framework rather than an absolute measure of industrial performance.

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Chapter 1 — Introduction

1.1 Background

Industrial cities play a fundamental role in supporting economic development and improving regional competitiveness. By providing integrated infrastructure, shared facilities, and organized industrial environments, they reduce operational costs, encourage industrial clustering, attract domestic and foreign investment, and stimulate innovation and employment opportunities. Consequently, industrial cities have become an important policy instrument for supporting sustainable industrialization and long-term economic growth (UNIDO, 2020; OECD, 2025).

Beyond their economic contribution, industrial cities strengthen regional development by improving the integration between industries, infrastructure, and supporting services. Well-planned industrial zones enhance resource utilization, increase productivity, create employment opportunities, and contribute to higher levels of regional and national economic performance. They also improve the business environment by facilitating investment and encouraging collaboration between industries, research institutions, and government agencies (Khalaf & Al-Jubouri, 2015; Ezzeddin, 2025).

In Saudi Arabia, industrial development represents one of the main pillars of Vision 2030, which aims to diversify the national economy and strengthen the industrial and logistics sectors. Through initiatives such as the National Industrial Development and Logistics Program (NIDLP), the Kingdom has invested extensively in industrial infrastructure, logistics networks, and industrial cities to enhance economic competitiveness and increase private sector participation. These initiatives have accelerated industrial expansion while improving connectivity between production centers and transportation networks (NIDLP, 2020).

The Eastern Province constitutes the country's largest industrial hub due to its strategic location, advanced transportation infrastructure, major seaports, railway connections, airports, and abundant energy resources. Within this region, the industrial cities of Dammam have experienced continuous expansion over the last two decades, making them among the most significant industrial centers in Saudi Arabia. Their strategic importance makes them an appropriate case study for evaluating industrial development and spatial readiness.

Evaluating industrial development requires more than conventional economic indicators because the performance and future expansion of industrial cities are strongly influenced by spatial factors such as accessibility, transportation networks, surrounding infrastructure, environmental constraints, and land suitability. Geographic Information Systems (GIS) provide an effective environment for integrating, managing, and analyzing these spatial datasets, allowing planners to better understand the spatial characteristics of industrial areas and support evidence-based planning. However, spatial analysis alone is insufficient for evaluating complex industrial systems that involve multiple spatial, environmental, and infrastructural criteria.

Furthermore, integrating GIS with Multi-Criteria Decision Analysis (MCDA) enables multiple spatial, environmental, and infrastructural criteria to be evaluated simultaneously, providing a more objective basis for industrial planning and decision-making. Such integrated approaches have become increasingly important for

supporting sustainable industrial development and improving the spatial planning of industrial cities (Malczewski & Rinner, 2015; Rikalovic et al., 2014).

1.2 Research Gap

Although Geographic Information Systems (GIS) have been widely adopted in industrial planning, previous studies have generally focused on individual spatial analyses, such as industrial site selection, accessibility, land suitability, industrial density, and temporal growth. While these approaches provide valuable insights into specific aspects of industrial development, they are typically applied as separate analyses rather than being integrated into a unified spatial decision-support framework (Rikalovic et al., 2014; Malczewski & Rinner, 2015).

Likewise, recent industrial readiness frameworks have expanded the evaluation of industrial performance by considering dimensions such as investment readiness, Industry 4.0 adoption, organizational capability, sustainability, and regulatory preparedness. However, these frameworks primarily emphasize managerial, technological, and institutional dimensions, with limited consideration of spatial characteristics that directly influence industrial performance, including accessibility, logistics connectivity, industrial clustering, and proximity to strategic infrastructure (WEF, 2022; Musyarofah et al., 2025).

Consequently, a gap remains between GIS-based spatial analytical approaches and industrial readiness assessment. Existing studies rarely integrate multiple GIS-derived spatial indicators into a single framework capable of evaluating the overall development readiness of industrial cities. Addressing this limitation requires an integrated approach that transforms diverse spatial analyses into standardized evaluation indicators, enabling comprehensive comparison and evidence-based decision-making.

The key gap identified in the literature is not the absence of individual GIS techniques, but the limited integration of these techniques into a unified industrial development readiness framework. No integrated GIS-based framework was identified that combines Industrial Density, Logistics Accessibility, Industrial Growth, Industrial Expansion Potential, and Industrial Specialization into a single comparative industrial development readiness assessment.

To address this gap, this research proposes the Industrial Development Readiness Model (IDRM), a GIS-based spatial decision-support framework that integrates multiple spatial analyses into a unified readiness assessment. The proposed model converts diverse GIS outputs into standardized spatial indicators, enabling systematic comparison among industrial cities, identification of development priorities, and support for sustainable industrial planning and investment decision-making.

1.3 Research Aim and Objectives

Research Aim

The primary aim of this research is to develop a GIS-based spatial decision-support framework, referred to as the Industrial Development Readiness Model (IDRM), for assessing and comparing the development readiness of industrial cities through the integration of multiple spatial indicators into a standardized evaluation framework.

Research Objectives

To achieve this aim, the study pursues the following objectives:

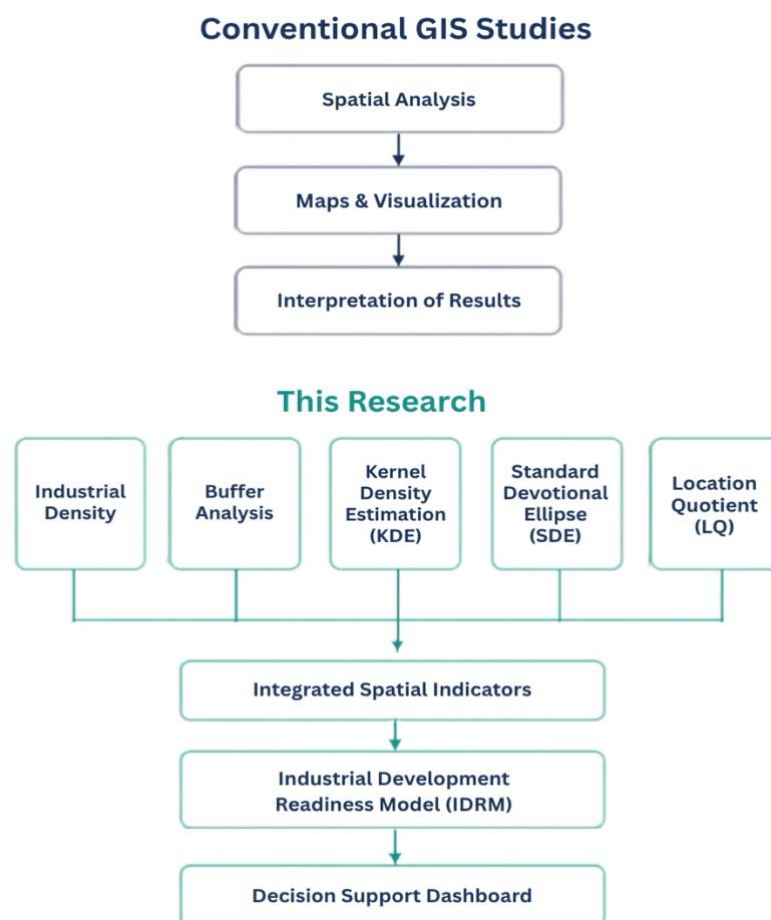
1. To analyze industrial density and spatial concentration patterns using Industrial Density Analysis and Kernel Density Estimation (KDE).
2. To analyze the spatial distribution, industrial density, and sectoral specialization of manufacturing activities.
3. To evaluate the accessibility of industrial cities to major logistics infrastructure, including transportation networks, seaports, airports, and strategic facilities.
4. To develop standardized GIS-derived spatial indicators representing key dimensions of industrial development readiness.
5. To construct the Industrial Development Readiness Model (IDRM) by integrating the developed indicators into a unified evaluation framework.
6. To compare the development readiness of the industrial cities using the proposed IDRM and identify their relative strengths and development priorities.
7. To demonstrate the applicability of the proposed framework for supporting evidence-based industrial planning, infrastructure prioritization, and strategic decision-making.

1.4 Project Contributions

This research contributes to the fields of industrial planning, spatial analysis, and GIS-based decision support through the following methodological and practical contributions:

- Developing the Industrial Development Readiness Model (IDRM), a GIS-based spatial decision-support framework for assessing the development readiness of industrial cities.
- Integrating multiple spatial analyses—including temporal change analysis, industrial density, sectoral specialization, accessibility, Kernel Density Estimation (KDE), and Standard Deviational Ellipse (SDE)—into a unified evaluation framework rather than treating them as independent analyses.
- Transforming diverse GIS outputs into standardized readiness indicators, enabling objective comparison between industrial cities using consistent evaluation criteria.
- Providing a transparent, repeatable, and evidence-based assessment methodology that reduces subjectivity in evaluating industrial development readiness.
- Supporting industrial planning and decision-making by identifying spatial strengths, development priorities, and infrastructure needs.
- Providing a transferable framework that can be adapted and applied to other industrial cities or regions with similar spatial datasets and planning objectives.

Figure 1.1. Research Contribution Framework



1.5 Report Structure

This report is organized into seven chapters that collectively present the development and application of the proposed Industrial Development Readiness Model (IDRM), from the research background to the final assessment of industrial cities using a GIS-based spatial decision-support framework.

Chapter 1 introduces the research background, identifies the research gap, presents the research aim and objectives, highlights the study contributions, and outlines the overall structure of the report.

Chapter 2 reviews the relevant literature on industrial development, Geographic Information Systems (GIS), Multi-Criteria Decision Analysis (MCDA), spatial decision-support systems, industrial readiness assessment, and the conceptual foundation of industrial development readiness. This chapter establishes the theoretical basis of the proposed framework.

Chapter 3 describes the study area and the industrial geospatial database, including data collection, database development, preprocessing, and data quality assessment.

Chapter 4 presents the research methodology, including the overall analytical workflow, GIS analytical techniques, and the methodological framework used to develop the proposed Industrial Development Readiness Model (IDRM).

Chapter 5 presents the results of the GIS-based spatial analyses, including macro-level spatial analysis, logistics accessibility analysis, micro-level spatial analysis, and comparative spatial analysis. The chapter concludes by synthesizing the principal spatial findings that provide the analytical foundation for constructing the Industrial Development Readiness Model (IDRM).

Chapter 6 presents the Industrial Development Readiness Model (IDRM), including the development of spatial indicators, indicator standardization, weighting procedures, model formulation, industrial readiness assessment, and the decision-support dashboard.

Chapter 7 summarizes the principal findings, discusses the practical implications of the proposed framework, identifies the study limitations, and provides recommendations for future research and the continued development of GIS-based industrial readiness assessment.

Chapter 2 — Literature Review and Theoretical Foundation

2.1 Industrial Cities

Industrial cities are founded on the principle of agglomeration, whereby firms benefit from locating close to one another. By sharing infrastructure, transportation networks, utilities, and supporting services, they reduce production and transaction costs while improving operational efficiency. Consequently, industrial cities have become an important policy instrument for promoting industrialization, economic diversification, and long-term structural transformation (UNIDO, 2019). Their role extends beyond reducing costs by strengthening both forward and backward industrial linkages, connecting suppliers, manufacturers, and markets within an integrated production system that facilitates technology transfer, attracts foreign direct investment (FDI), and supports industrial upgrading (UNIDO, 2019).

Beyond improving manufacturing performance, industrial cities also contribute to broader regional development. Instead of concentrating economic activity within a limited number of metropolitan areas, they enable different regions to capitalize on their own resources, industrial capabilities, and competitive advantages. According to the OECD (2025), place-based industrial policies generally outperform uniform national approaches because they are designed around local assets, specialized industries, and regional knowledge. This principle is reflected in initiatives such as the United Kingdom's Local Industrial Strategies, which align industrial policy with regional innovation systems, workforce development, and local competitive strengths (OECD, 2025).

Research from different national contexts further demonstrates the close relationship between industrial development and economic performance. Khalaf and Al-Jubouri (2015) found that well-planned industrial zones encourage private investment by providing reliable infrastructure and an organized business environment. Likewise, Ezzeddin (2025) argues that industrial cities contribute not only to economic growth and employment generation but also to technological advancement, research collaboration, and the diffusion of modern manufacturing practices. Taken together, these findings suggest that industrial cities should be viewed as integrated development platforms rather than simply locations where manufacturing activities are concentrated.

The theoretical foundation of these advantages lies in agglomeration economies. Meyer (2000) explains that firms located within the same industrial cluster benefit from shared infrastructure, specialized labor markets, supplier networks, and knowledge exchange, all of which improve productivity while reducing production costs. These benefits, however, are not unlimited. As industrial concentration increases, congestion, environmental pressures, and competition for land and resources may gradually outweigh the advantages of clustering. Within this context, Meyer further distinguishes between industries that are primarily influenced by transportation costs and those driven by agglomeration economies. Resource-based industries, such as steel and petroleum refining, tend to locate near raw materials or major transportation infrastructure, whereas technology-intensive and light manufacturing industries are more strongly attracted to skilled labor, specialized suppliers, and complementary industries where knowledge exchange plays a greater role (Meyer, 2000).

This perspective is further reinforced by Henderson, Kuncoro, and Turner (1992), who demonstrate that localized knowledge spillovers and accumulated industrial

experience encourage firms to remain clustered over long periods. Their findings also indicate that cities with more diversified economic structures provide greater opportunities for innovation and long-term industrial adaptability than those relying on a narrow industrial base. Therefore, sustainable industrial competitiveness depends not only on specialization but also on maintaining sufficient economic diversity to support continuous learning, knowledge exchange, and resilience to changing market conditions.

2.2 Industrial Development in Saudi Arabia and the Selection of Dammam

Industrial development has become a central component of Saudi Arabia's economic diversification strategy under Vision 2030. As the Kingdom seeks to reduce its dependence on oil revenues, increasing emphasis has been placed on expanding manufacturing industries, strengthening logistics services, and improving industrial infrastructure. These priorities are reflected in the National Industrial Development and Logistics Program (NIDLP), which aims to enhance industrial competitiveness by integrating production, transportation, logistics, and investment across the country (NIDLP, 2019).

Saudi Arabia's industrial strategy recognizes that different regions possess distinct competitive advantages. Rather than adopting a uniform development model nationwide, industrial policies are designed to capitalize on each region's geographic location, natural resources, existing industries, and infrastructure. Within this framework, the Eastern Province occupies a particularly important position because of its concentration of energy resources, major seaports, railway connections, airports, and extensive logistics networks. These characteristics have established the region as the Kingdom's leading industrial hub and one of the principal drivers of Saudi Arabia's industrial expansion (NIDLP, 2019).

This national strategy is reflected in the continued expansion of industrial cities managed by the Saudi Authority for Industrial Cities and Technology Zones (MODON). According to the MODON Annual Report (2024), industrial land allocation, factory numbers, and private-sector investment have increased steadily across the Kingdom, reflecting growing demand for organized industrial environments supported by modern infrastructure and public services. These trends demonstrate the increasingly important role of industrial cities in supporting economic growth while providing suitable environments for industrial production and long-term investment (MODON, 2024).

Among these industrial cities, Dammam represents one of the most significant examples of industrial development in Saudi Arabia. Located in the Eastern Province, the industrial cities of Dammam benefit from direct access to major transportation corridors, King Abdulaziz Port, King Fahd International Airport, railway connections, and the Kingdom's largest concentration of oil and petrochemical industries. These strategic advantages have supported sustained industrial growth over the past two decades and strengthened Dammam's position as one of Saudi Arabia's leading manufacturing and logistics centers.

These characteristics make Dammam an appropriate case study for examining industrial development from a spatial perspective. Its diverse manufacturing base,

sustained industrial expansion, and close integration with regional transportation and logistics infrastructure provide a suitable environment for investigating how spatial characteristics influence industrial development readiness. Equally important, the availability of reliable industrial and geospatial datasets enables spatial indicators to be developed and analyzed consistently, making Dammam particularly well suited for the GIS-based assessment undertaken in this research.

Understanding how these spatial characteristics influence industrial development requires analytical tools capable of integrating multiple geographic datasets within a unified analytical framework. Geographic Information Systems (GIS) provide this capability by supporting the collection, management, visualization, and analysis of spatial information. The following section therefore examines the role of GIS in industrial planning and explains why it has become a fundamental component of modern spatial decision-support systems.

2.3 Geographic Information Systems (GIS) in Industrial Planning

Geographic Information Systems (GIS) have become an essential component of industrial planning because many planning decisions are inherently spatial. The selection of industrial sites, evaluation of accessibility, assessment of transportation networks, identification of environmental constraints, and analysis of development opportunities all depend on geographic relationships that cannot be fully understood through conventional statistical methods alone. By integrating spatial and non-spatial information within a unified environment, GIS enables planners to visualize, manage, and analyze complex datasets, supporting more informed and transparent planning decisions.

One of the principal strengths of GIS is its ability to integrate transportation networks, land use, industrial facilities, environmental constraints, utilities, and demographic information within a single spatial framework. Examining these datasets together reveals spatial relationships that often remain hidden when each layer is analyzed independently, allowing industrial areas to be evaluated from multiple planning perspectives rather than through isolated criteria. As a result, GIS provides a more comprehensive understanding of the spatial conditions that influence industrial development.

Beyond map production, GIS offers a wide range of analytical techniques that support industrial planning and policy development. Spatial distribution analysis, proximity analysis, density estimation, accessibility assessment, suitability modelling, and change detection enable planners to identify development patterns, evaluate infrastructure relationships, and recognize opportunities and constraints affecting industrial growth. These analytical capabilities transform large volumes of geographic information into meaningful spatial evidence that supports strategic planning and long-term industrial development.

The contribution of GIS to industrial planning has been demonstrated in numerous studies. Rikalovic et al. (2014) developed a GIS-based framework that combined spatial

analysis with multi-criteria evaluation to support industrial site selection. Their approach incorporated screening criteria such as road accessibility, environmental constraints, and proximity to water resources, illustrating how GIS can improve the transparency and consistency of site evaluation. Similarly, Sarath et al. (2018) integrated GIS with the Analytic Hierarchy Process (AHP) to evaluate industrial land suitability, demonstrating that weighted spatial criteria provide a more objective basis for planning decisions than individual indicators alone. More recently, Chumaidiyah et al. (2023) integrated GIS and AHP within a web-based decision-support application to evaluate industrial site suitability using multiple spatial variables, including land use, slope, road accessibility, proximity to rivers, and public facilities. Their findings further demonstrate that GIS can systematically combine diverse spatial criteria within a consistent and reproducible evaluation framework.

The expanding role of GIS has therefore transformed it from a system primarily used for storing and displaying spatial information into an analytical platform that supports evidence-based decision-making. According to Malczewski and Rinner (2015), many planning problems are only partially structured because they involve both measurable spatial information and expert judgment. In such situations, GIS provides the analytical environment for processing spatial data, while complementary decision-support methods assist planners in evaluating competing alternatives through explicit and transparent criteria.

Despite these advances, most GIS applications in industrial planning remain focused on individual analyses, such as accessibility assessment, industrial density, temporal change, land suitability, or site selection. Although these approaches provide valuable insights into specific planning questions, they rarely integrate multiple spatial analyses into a comprehensive evaluation of industrial development readiness. This limitation highlights the need for analytical frameworks capable of combining multiple spatial indicators within a unified assessment, allowing industrial cities to be evaluated from a broader strategic perspective rather than through isolated spatial outputs.

While GIS provides a powerful environment for spatial analysis, planning decisions frequently involve multiple and sometimes competing objectives that cannot be resolved through spatial analysis alone. Spatial criteria must also be standardized, weighted, and evaluated according to their relative importance. Consequently, GIS is increasingly integrated with Multi-Criteria Decision Analysis (MCDA), which provides a structured methodology for combining multiple criteria within a transparent decision-support framework. The following section therefore reviews the principles of MCDA and explains how it complements GIS in addressing complex industrial planning problems.

2.4 Multi-Criteria Decision Analysis (MCDA)

Industrial planning decisions rarely depend on a single factor. Decisions such as evaluating industrial development, selecting suitable locations, or prioritizing future investment require multiple criteria to be considered simultaneously. Economic, environmental, infrastructural, and spatial considerations often interact with one another, and improvements in one aspect may involve trade-offs in another. Consequently, industrial planning requires systematic approaches capable of evaluating diverse criteria in a transparent and consistent manner.

Multi-Criteria Decision Analysis (MCDA) provides such a framework by enabling different sources of information to be organized, standardized, weighted, and integrated into a structured decision-making process. Rather than relying solely on individual indicators or subjective judgment, MCDA supports the comparison of alternative solutions based on multiple evaluation criteria while making the decision-making process more transparent and easier to justify (Malczewski & Rinner, 2015).

The integration of GIS and MCDA has become increasingly important in spatial planning because the two approaches serve complementary roles. GIS provides the analytical environment for managing and processing spatial information, whereas MCDA evaluates the relative importance of different criteria and combines them into a coherent decision-making framework. Together, they enable planners to address complex spatial problems that cannot be adequately resolved through individual spatial analyses alone (Malczewski & Rinner, 2015).

Among the various MCDA techniques, the Analytic Hierarchy Process (AHP) remains one of the most widely applied methods in spatial planning. Developed by Saaty (1980), AHP determines the relative importance of decision criteria through pairwise comparisons, allowing qualitative expert judgments to be transformed into quantitative weights. The method also incorporates a consistency ratio that evaluates the logical coherence of these judgments, making AHP particularly suitable for planning problems involving multiple and competing criteria.

The practical benefits of combining GIS and AHP have been demonstrated in several industrial planning studies. Rikalovic et al. (2014) integrated AHP within a GIS-based framework to evaluate industrial site suitability using multiple spatial criteria during the screening and evaluation stages of site selection. Likewise, Sarath et al. (2018) applied GIS-AHP to industrial land suitability assessment and reported that weighted spatial criteria produced more objective and reliable planning outcomes than evaluating individual indicators separately. More recently, Chumaidiyah et al. (2023) employed AHP to assign weights to six spatial variables—including land use, slope, accessibility, soil type, proximity to rivers, and public facilities—within a web-based GIS application. Their study further demonstrated that combining weighted spatial indicators within an integrated framework provides a practical and transparent approach for evaluating industrial development alternatives.

Despite the growing adoption of GIS-MCDA, most existing applications remain focused on industrial site selection or land suitability assessment, with relatively limited attention given to evaluating the overall development readiness of existing industrial cities. Consequently, few studies integrate multiple spatial analyses—such as accessibility, industrial clustering, infrastructure, and environmental constraints—

within a single readiness assessment framework. This limitation indicates the need for broader evaluation models capable of supporting strategic industrial planning beyond conventional suitability mapping.

Building on these insights, the present research treats GIS and MCDA as complementary components of a broader spatial evaluation framework. GIS provides the analytical basis for deriving spatial indicators, whereas MCDA offers a systematic methodology for standardizing, weighting, and integrating those indicators into a consistent assessment process. Together, these approaches form the methodological foundation for evaluating industrial development readiness in a transparent and evidence-based manner.

2.5 Spatial Decision Support Systems (SDSS)

Although Geographic Information Systems (GIS) provide powerful capabilities for managing and analyzing spatial information, effective industrial planning requires more than spatial analysis alone. Planning decisions often involve balancing multiple criteria, including accessibility, infrastructure availability, environmental constraints, investment attractiveness, and land-use efficiency. Consequently, decision-makers require analytical environments capable of integrating spatial information with structured evaluation methods.

Spatial Decision Support Systems (SDSS) were developed to meet this need by combining GIS with decision-analysis techniques within a unified framework. According to Chakhar and Mousseau (2008), SDSS extends conventional GIS by supporting the evaluation of alternative scenarios, integrating multiple sources of information, and producing transparent and evidence-based recommendations for decision-makers.

Previous research has demonstrated the practical value of SDSS in industrial planning. Eldrandaly et al. (2003) developed an SDSS that integrated GIS, the Analytic Hierarchy Process (AHP), and an expert system to support industrial site selection through spatial screening and multi-criteria evaluation. More recently, Mallick et al. (2022) combined GIS with Fuzzy AHP to evaluate development suitability under uncertainty, while Jelokhani-Niaraki et al. (2023) proposed a collaborative web-based SDSS that enabled multiple stakeholders to evaluate spatial alternatives using shared geographic information. Similarly, Chumaidiyah et al. (2023) demonstrated that integrating GIS and AHP within a web-based platform improves the consistency, transparency, and reproducibility of industrial spatial evaluation.

These studies indicate that SDSS has evolved beyond traditional site-selection applications into a comprehensive spatial planning environment capable of integrating GIS, multi-criteria evaluation, expert knowledge, and collaborative decision-making. Such integration enables planners to evaluate complex industrial development problems from multiple spatial perspectives rather than relying on individual analytical outputs.

Despite these advances, most SDSS applications continue to focus primarily on industrial site selection or land suitability assessment. Comparatively fewer studies have employed SDSS principles to evaluate the overall development readiness of existing industrial cities through the integration of multiple GIS-derived indicators. This

limitation highlights the need for broader spatial evaluation frameworks capable of supporting strategic industrial planning beyond conventional suitability mapping.

Building upon these principles, the Industrial Development Readiness Model (IDRM) proposed in this research adopts the fundamental concepts of SDSS while addressing a different planning objective. Instead of identifying suitable locations for future industrial development, the framework integrates multiple GIS-derived spatial indicators into a standardized readiness assessment, enabling systematic comparison among existing industrial cities and supporting evidence-based industrial planning and decision-making.

2.6 Summary and Research Positioning

The literature reviewed in this chapter demonstrates that industrial development is influenced by the interaction of economic, spatial, and institutional factors. Industrial cities enhance regional competitiveness by concentrating infrastructure, industries, and supporting services within organized environments, while national industrial policies provide the strategic direction for sustainable industrial growth. In Saudi Arabia, Vision 2030 and the National Industrial Development and Logistics Program (NIDLP) have accelerated industrial expansion and strengthened logistics infrastructure, particularly within the Eastern Province, where the industrial cities of Dammam represent one of the Kingdom's principal manufacturing and logistics hubs.

The review also confirms that Geographic Information Systems (GIS), Multi-Criteria Decision Analysis (MCDA), and Spatial Decision Support Systems (SDSS) have become fundamental components of contemporary industrial planning. Together, these approaches provide an effective analytical environment for managing spatial information, integrating multiple evaluation criteria, and supporting transparent, evidence-based decision-making.

Despite these methodological advances, previous studies have generally examined individual aspects of industrial planning—such as accessibility, industrial density, land suitability, or site selection—rather than integrating multiple spatial analyses within a unified framework capable of evaluating the overall development readiness of existing industrial cities. Likewise, existing industrial readiness frameworks have primarily emphasized technological, organizational, or institutional dimensions while giving comparatively limited attention to spatial characteristics.

To address this gap, this research proposes the Industrial Development Readiness Model (IDRM), a GIS-based spatial decision-support framework that integrates multiple spatial analyses into standardized readiness indicators. By combining GIS, MCDA, and SDSS principles within a unified evaluation framework, the proposed model enables systematic comparison among industrial cities while providing a transparent and evidence-based assessment of industrial development readiness.

Accordingly, the following chapter introduces the study area, describes the industrial geospatial database, and presents the data preparation procedures that support the development and implementation of the proposed Industrial Development Readiness Model (IDRM).

Chapter 3 — Study Area and Geospatial Database

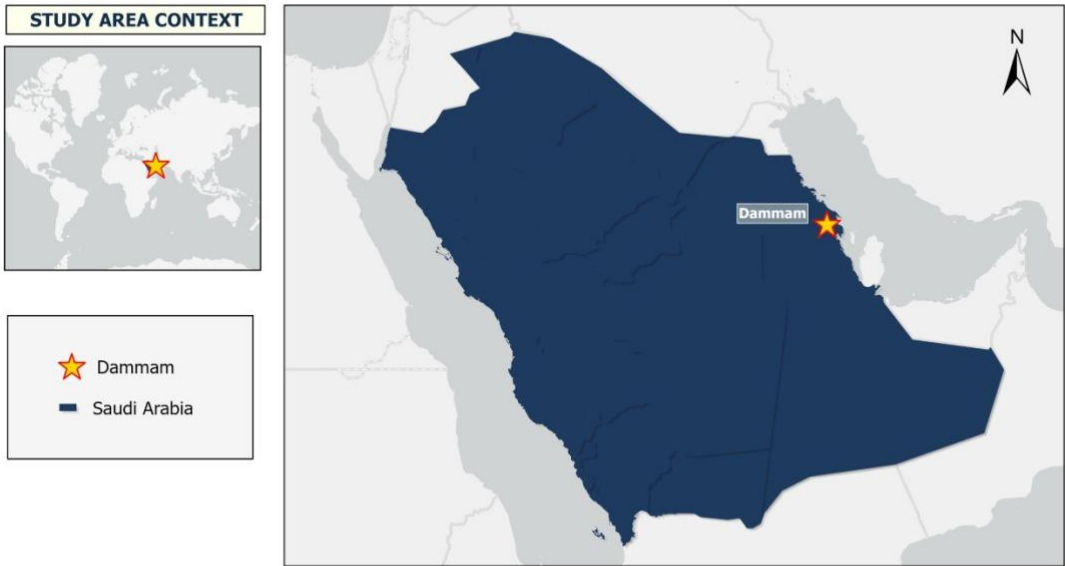
3.1 Study Area

D1 First Industrial City	D2 Second Industrial City	D3 Third Industrial City
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The study focuses on the three industrial cities located in Dammam, Eastern Province, Saudi Arabia: the First Industrial City (D1), the Second Industrial City (D2), and the Third Industrial City (D3). Together, these industrial cities represent one of the Kingdom's largest industrial concentrations and form a major component of Saudi Arabia's manufacturing and logistics sector under Vision 2030 and the National Industrial Development and Logistics Program (NIDLP).

Situated within the Eastern Province, the industrial cities benefit from a strategic geographic location supported by direct access to regional and international transportation infrastructure. Their proximity to King Abdulaziz Port, King Fahd International Airport, the national railway network, and major highway corridors facilitates the efficient movement of raw materials, manufactured products, and industrial services. Although some supporting logistics facilities included in this research are located outside the Dammam municipal boundary, they were incorporated because they function as regional logistics assets that directly influence the accessibility and operational performance of the industrial cities.

LOCATION OF DAMMAM WITHIN SAUDI ARABIA

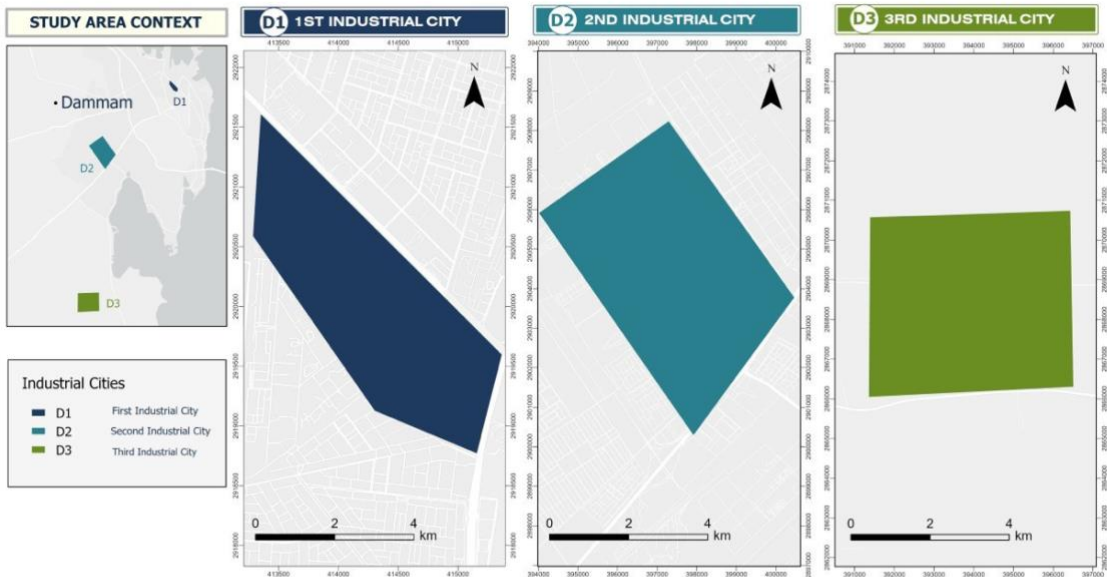


Geographic Reference : WGS 1984

Prepared by: **Aryam Alsaidi**

Figure 3.1.

DAMMAM INDUSTRIAL CITIES (1st, 2nd & 3rd)



Coordinate System: WGS 1984 UTM Zone 39N

Prepared by: **Aryam Alsaidi**

Figure 3.2

The geographic location of the study area is presented in Figures 3.1 and 3.2. Figure 3.1 identifies the location of Dammam within Saudi Arabia, while Figure 3.2 illustrates the spatial distribution of the three industrial cities across the study area.

The three industrial cities differ considerably in terms of establishment period, development stage, industrial composition, spatial expansion, and infrastructure characteristics. These differences provide an appropriate environment for comparative spatial analysis and enable industrial development to be examined from both spatial and temporal perspectives over the period 2000–2025.

Dammam was selected as the study area for three principal reasons. First, it represents one of the largest industrial clusters in Saudi Arabia and plays a central role in the Kingdom's manufacturing and logistics sectors. Second, its industrial cities have experienced continuous expansion during the last two decades, providing a suitable basis for investigating long-term industrial development patterns. Third, the availability of reliable industrial records, official statistics, and geospatial datasets enabled the development of a comprehensive industrial geodatabase and the derivation of consistent GIS-based indicators throughout this research.

3.2 Data Sources and Collection

This study relied on multiple geospatial datasets collected from official governmental sources, publicly available geospatial repositories, and manually compiled industrial records. These datasets were organized and integrated into a unified GIS geodatabase to support the macro-level, micro-level, comparative, and logistics analyses conducted throughout this research. The principal datasets used in the study, together with their sources and analytical purposes, are summarized in Table 3.2.

The geospatial datasets used in this research were collected from multiple official and publicly available sources. Table 3.2 summarizes the datasets, their sources, and their analytical purposes.

Table 3.2. Geospatial Database and Sources

Level	Spatial Dataset	Source	Analytical Purpose
Macro	Industrial City Boundaries	MODON	Macro-level spatial analysis
Micro	Industrial Facilities Database	Author-compiled database	Factory-level spatial analysis
Logistics	Road & Railway Infrastructure	OpenStreetMap (OSM)	Logistics accessibility analysis
Logistics	Airport & Railway Station Infrastructure	Author-compiled dataset	Air & Railway transport accessibility analysis
Logistics	Port Infrastructure	Author-compiled dataset	Maritime logistics accessibility analysis
Logistics	Energy Infrastructure	Author-compiled dataset	Energy infrastructure accessibility analysis

Following data collection, the datasets were classified according to their spatial geometry to facilitate spatial analysis within the GIS environment. Polygon, line, and point feature classes were created to ensure consistency in data organization and to support the different GIS analyses performed throughout this study. The resulting spatial database structure is presented in Table 3.3.

Table 3.3. Spatial Database Structure

Spatial Layer	Geometry Type	Primary Use this Study
Industrial Cities	Polygon	Study area boundaries
Industrial Facilities	Point	Factory-level analysis
Major Roads	Line	Accessibility reference
Railway Lines	Line	Accessibility reference
Airport & Railway Stations	Point	Buffer accessibility analysis
Port	Point	Maritime accessibility analysis
Oil & Gas Facilities	Point	Industrial logistics support

The following section describes the procedures used to prepare, standardize, and validate the collected datasets before their integration into the analytical workflow.

3.3 Database Development and Data Preparation

A geodatabase was developed following a standardized data preparation workflow to ensure consistency, completeness, and compatibility for spatial analysis within ArcGIS Pro. Before analysis, all datasets were organized into separate feature classes within a unified geodatabase, where attribute names, data formats, coordinate systems, and classification schemes were standardized to facilitate efficient data management and subsequent GIS analyses.

The geodatabase contains three principal feature classes representing the Industrial Cities, Industrial Facilities, and Logistics Infrastructure. The Industrial Cities feature class was created using official MODON boundaries and attribute information, while the Industrial Facilities feature class was manually compiled by the author, with each factory record cross-checked against more than 200 official and publicly available sources for verification. Each factory location was individually verified, and its geographic coordinates were extracted using Google Earth Pro before being incorporated into the geodatabase. The verified records were subsequently reduced to a standardized, balanced research sample of 40 factories per industrial city (120 factories in total) for the micro-level and comparative analyses; this 120-factory sample must not be confused with the official MODON factory totals used in the macro-level density analysis.

The Logistics Infrastructure component was organized into five feature classes representing the road network, railway network, airports and railway stations, ports, and strategic oil and gas infrastructure. Road and railway datasets were obtained from OpenStreetMap (OSM) and processed in ArcGIS Pro, whereas the remaining logistics datasets were compiled from official national sources.

Figure 3.3. illustrates the conceptual organization of the integrated geodatabase developed for the Industrial Development Readiness Model (IDRM).



All datasets were projected to a common coordinate system (WGS 1984 UTM Zone 39N) to ensure positional consistency and improve the accuracy of distance-based analyses such as Buffer and Near operations.

Data quality was maintained throughout database development by verifying record completeness, removing duplicate entries, standardizing attribute values, and resolving missing information through cross-checking with official sources whenever possible. This process produced a reliable and internally consistent geodatabase suitable for all spatial analyses presented in this research.

3.4 Database Structure

The geodatabase was structured into three interconnected thematic components representing different levels of analysis within the Industrial Development Readiness Model (IDRM). Rather than storing all information in a single feature class, separate feature classes were created according to their analytical roles. This organization improves database management, reduces data redundancy, and facilitates efficient spatial processing and attribute management within ArcGIS Pro.

The Industrial Cities feature class stores macro-level information describing each industrial city, including its establishment year, development status, total number of factories, and industrial density. The Industrial Facilities feature class contains detailed records for the sampled factories used in the micro-level analyses, including industrial classification, temporal attributes, and geographic coordinates. The Logistics Infrastructure component consists of five feature classes representing the road network, railway network, airports and railway stations, ports, and strategic oil and gas infrastructure, all of which support the accessibility analyses.

Table 3.4 summarizes the principal attributes contained within each feature class.

Table 3.4 Database Attributes

Database	Primary Attributes
Industrial Cities	City Name, Establishment Year, Development Status , Total Factories, Industrial Density (Factories/km ²)
Industrial Facilities	Factory ID, Factory Name, Industry Type, Establishment Year, Industrial City (D1–D3), Temporal Status (2000, 2015, 2020, 2025), Latitude, Longitude
Logistics Infrastructure	Name, Infrastructure Type

The Industrial Facilities feature class was specifically designed to support both spatial and temporal analyses. Each factory record includes a unique identifier, industrial classification, establishment year, industrial city code, geographic coordinates, and four binary temporal fields (2000, 2015, 2020, and 2025) indicating the operational status of each facility during the study period. This structure enabled the temporal evolution of industrial development to be analyzed consistently across the 25-year study period. Industry types were standardized into nine manufacturing categories to ensure classification consistency and support comparative analyses, including temporal industrial development and industrial specialization using the Location Quotient (LQ).

3.5 Chapter Summary

This chapter introduced the study area and described the geospatial database developed for the Industrial Development Readiness Model (IDRM). It presented the data sources, database development procedures, data preparation workflow, and the overall database structure adopted for the study. Together, these datasets provide the spatial foundation required for the GIS analyses presented in the following chapter.

Chapter 4 — Research Methodology

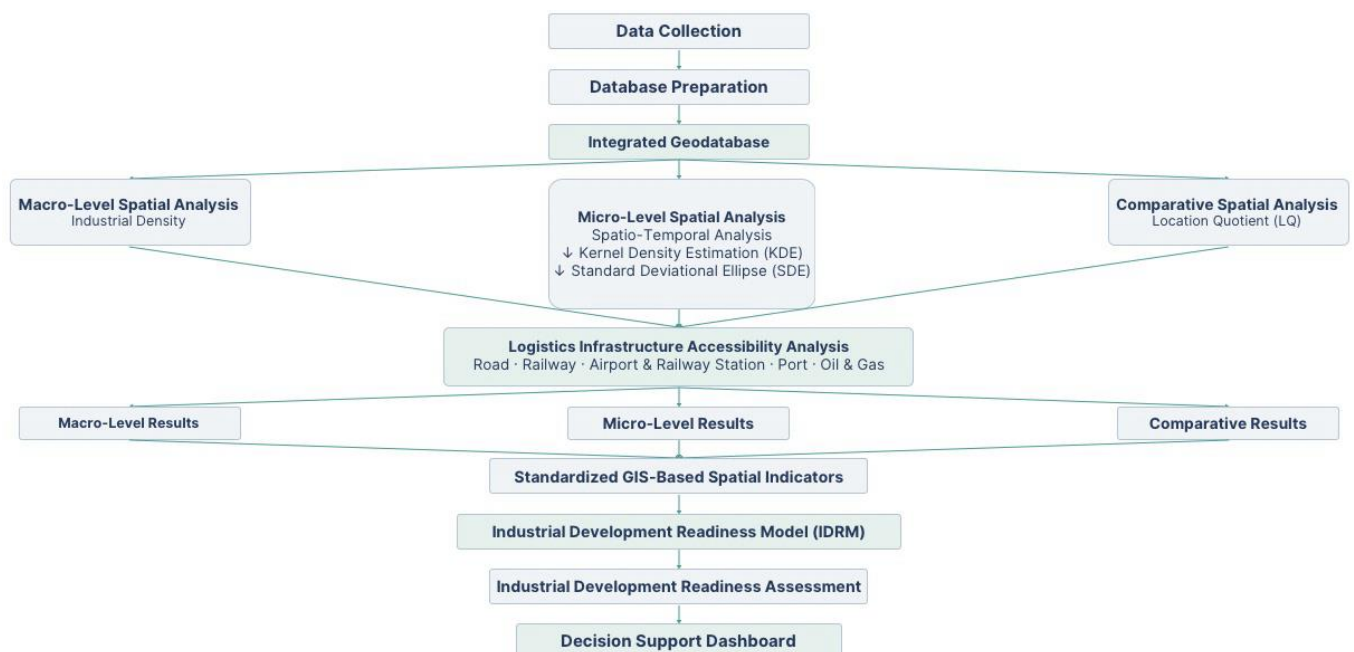
4.1 Overall Research Workflow

The methodological framework adopted in this study consists of a multi-level GIS-based spatial analysis process integrating macro-level, micro-level, and comparative analyses. Rather than relying on a single analytical technique, the framework combines multiple complementary spatial methods to investigate industrial development from different spatial perspectives before integrating the findings into the Industrial Development Readiness Model (IDRM).

The workflow begins with the integrated geodatabase developed in Chapter 3, which contains the Industrial Cities, Industrial Facilities, and Logistics Infrastructure feature classes. The analytical process is then organized into three complementary levels. The macro-level analysis evaluates industrial cities as spatial units through Industrial Density and Buffer analyses. The micro-level analysis investigates the spatial distribution and temporal evolution of individual industrial facilities using Spatio-Temporal Analysis, Kernel Density Estimation (KDE), and Standard Deviational Ellipse (SDE). The comparative level applies the Location Quotient (LQ) to evaluate industrial specialization and compare the industrial structure of the three industrial cities.

Finally, the outputs generated from the three analytical levels are transformed into standardized GIS-based spatial indicators, which collectively form the input variables of the Industrial Development Readiness Model (IDRM). The model integrates these indicators into a unified framework for assessing industrial development readiness and supporting evidence-based spatial decision-making.

Figure 4.1 Overall methodological workflow for the development of the Industrial Development Readiness Model (IDRM).



4.2 Macro-Level Spatial Analysis

Macro-level spatial analysis was conducted to evaluate industrial development at the scale of industrial cities rather than individual industrial facilities. At this analytical level, each industrial city was treated as a single spatial unit, allowing comparisons of their overall development characteristics, industrial concentration, and accessibility to strategic logistics infrastructure. This level of analysis provides a broad spatial perspective that complements the detailed facility-level investigations presented in the subsequent micro-level analyses.

Two GIS-based analyses were performed within the macro-level framework. Industrial Density was calculated to compare the intensity of industrial development among the three industrial cities by relating the total number of factories to the area of each industrial city. Buffer Analysis was subsequently applied to evaluate the spatial accessibility of industrial facilities to major logistics infrastructure, including the road network, railway station, airport, port, and oil and gas infrastructure. Together, these analyses generated macro-scale spatial indicators that contribute to the Industrial Development Readiness Model (IDRM).

4.2.1 Industrial Density Analysis

Industrial density is one of the most widely used indicators for evaluating the intensity of industrial development within a defined spatial unit. Rather than relying solely on the total number of industrial facilities, density standardizes industrial concentration by considering the size of the study area, thereby enabling objective comparisons between industrial zones of different spatial extents. This normalization reduces potential bias associated with differences in city size and provides a more representative measure of industrial concentration.

The industrial density for each industrial city was calculated as the ratio between the total number of industrial facilities and the total area of the industrial city, expressed as factories per square kilometre (factories/km²). This indicator provides a standardized measure of industrial concentration that facilitates direct comparison among the three industrial cities despite differences in their spatial extent.

Equation 4.1. Industrial Density Calculation

$$\text{Industrial Density} = \frac{\text{Total Number of Factories}}{\text{Industrial City Area (km}^2\text{)}}$$

In this study, the total number of factories and the official boundaries of each industrial city were obtained from the MODON database. Industrial city areas were derived from the official MODON spatial boundaries, allowing industrial density values to be calculated consistently for all three industrial cities. The resulting indicator was subsequently used to compare the relative intensity of industrial development and to provide one of the macro-level spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

4.2.2 Buffer Analysis

Buffer analysis was employed as the primary GIS proximity analysis technique to evaluate logistics infrastructure accessibility. It is one of the most fundamental vector-based spatial analysis methods in Geographic Information Systems (GIS), widely used to examine spatial relationships by creating zones at specified distances around geographic features. Rather than measuring travel distance or travel time through transportation networks, buffer analysis evaluates accessibility based on Euclidean distance, providing an efficient approach for identifying areas located within predefined zones of influence (Ma et al., 2018).

In industrial planning, buffer analysis is commonly applied to assess the spatial relationship between industrial activities and supporting infrastructure. By defining distance-based accessibility zones around strategic logistics facilities, planners can evaluate the relative spatial advantage of industrial locations with respect to transportation networks, ports, airports, and energy infrastructure. Such analyses provide valuable insights into the spatial distribution of industrial accessibility and the logistical support available for industrial development.

In this study, the Multiple Ring Buffer Analysis tool was applied to evaluate the accessibility of industrial facilities to five categories of logistics infrastructure: the road network, railway stations, airports, ports, and oil and gas infrastructure. Three concentric buffer zones (5 km, 10 km, and 15 km) were generated for each logistics component to represent increasing levels of spatial accessibility. Industrial facilities located within the 5 km buffer were considered to have the highest level of spatial accessibility, whereas facilities located within the 10 km and 15 km buffer zones represented progressively lower levels of accessibility.

The resulting accessibility measures were subsequently transformed into standardized GIS-based accessibility indicators, forming one of the logistics dimensions incorporated into the Industrial Development Readiness Model (IDRM). These indicators are presented and interpreted in Chapter 5 as part of the Logistics Infrastructure Accessibility analysis.

Figure 4.2. Conceptual illustration of the Multiple Ring Buffer analysis adopted in this study.



4.3 Micro-Level Spatial Analysis

Micro-level spatial analysis focuses on the spatial characteristics and temporal evolution of individual industrial facilities rather than industrial cities as aggregated spatial units. While macro-level analysis evaluates industrial development from a city-wide perspective, micro-level analysis investigates the distribution, concentration, and directional patterns of individual factories, providing a more detailed understanding of industrial development processes. This analytical level enables localized spatial patterns to be identified that may not be evident from aggregated city-level indicators.

In this study, micro-level analysis was conducted using three complementary GIS techniques: Spatio-Temporal Analysis, Kernel Density Estimation (KDE), and Standard Deviational Ellipse (SDE). Spatio-temporal analysis was employed to examine the evolution of industrial facilities throughout the study period, while Kernel Density Estimation was used to identify areas of industrial concentration and spatial intensity. Standard Deviational Ellipse analysis was subsequently applied to evaluate the overall directional trend and spatial dispersion of industrial development.

Together, these analytical techniques provide a comprehensive assessment of industrial development at the facility level and generate micro-level spatial indicators that contribute to the Industrial Development Readiness Model (IDRM).

4.3.1 Spatio-Temporal Analysis

Spatio-temporal analysis integrates spatial and temporal dimensions to investigate how geographic phenomena evolve over time. Within a GIS environment, this approach enables the visualization and comparison of spatial changes across multiple time periods, revealing patterns of growth, expansion, and distribution that cannot be identified through purely spatial or purely temporal analyses (Budnicka-Kosior et al., 2025).

In industrial development studies, spatio-temporal analysis is widely applied to examine the evolution of industrial facilities and to identify changes in their spatial distribution throughout different stages of development. By comparing successive temporal observations, it becomes possible to identify periods of industrial expansion, detect emerging development patterns, and evaluate the long-term dynamics of industrial growth.

In this study, spatio-temporal analysis was performed using four reference years (2000, 2015, 2020, and 2025). Each industrial facility within the Industrial Facilities Database was assigned a binary temporal status indicating whether the facility existed during each reference year. These temporal attributes enabled the generation of sequential spatial datasets representing the progressive development of industrial facilities across the 25-year study period.

The resulting temporal maps provide a visual representation of industrial expansion and support the interpretation of the subsequent Kernel Density Estimation (KDE) and Standard Deviational Ellipse (SDE) analyses. Furthermore, the identified temporal development patterns support the interpretation of the micro-level spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

4.3.2 Kernel Density Estimation (KDE)

Kernel Density Estimation (KDE) is a widely used spatial statistical technique that transforms discrete point features into a continuous density surface, enabling the identification of areas with relatively high or low concentrations of spatial phenomena. Unlike simple point mapping, KDE estimates the intensity of spatial distribution by assigning greater influence to nearby observations than to more distant ones, thereby providing a smoothed representation of spatial clustering (Mora-Garcia et al., 2015).

In industrial studies, KDE is commonly applied to identify industrial concentration patterns, detect emerging clusters, and visualize the spatial intensity of industrial activities. The resulting density surfaces provide an effective means of highlighting areas where industrial facilities are spatially concentrated, supporting the interpretation of industrial development patterns that may not be readily apparent from individual point distributions alone.

Previous research has demonstrated that the selection of the search radius (bandwidth) has a greater influence on KDE results than the choice of the kernel function itself. An appropriately selected bandwidth produces meaningful density surfaces that balance local detail with overall spatial trends, whereas excessively small or large bandwidths may either fragment or oversmooth the resulting density patterns (Mora-Garcia et al., 2015).

In this study, Kernel Density Estimation was performed separately for each industrial city using the industrial facilities dataset. The search radius (bandwidth) was manually determined according to the spatial extent of each industrial city and the distribution of industrial facilities to ensure that the resulting density surfaces accurately represented local industrial concentration patterns. The generated density maps were subsequently used to identify industrial hotspots and support the development of the micro-level spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

4.3.3 Standard Deviational Ellipse (SDE)

Standard Deviational Ellipse (SDE) is a spatial statistical technique used to summarize the directional characteristics and geographic dispersion of spatial point distributions. Unlike density-based methods, SDE provides a simplified representation of the overall spatial orientation, central tendency, and distributional spread of geographic features by generating an ellipse that reflects the dominant direction of spatial development (Zhao et al., 2023).

In industrial spatial analysis, SDE is widely applied to investigate the directional evolution of industrial development and to identify long-term spatial trends. By examining changes in the orientation and dispersion of industrial facilities over successive time periods, the method provides valuable insights into the dominant direction of industrial expansion and the degree of spatial concentration or dispersion.

In this study, Standard Deviational Ellipse analysis was applied separately to the industrial facilities corresponding to the four reference years (2000, 2015, 2020, and 2025). Individual ellipses were generated for each temporal dataset to evaluate changes in the spatial orientation and geographic distribution of industrial development

throughout the study period. Comparing these ellipses enabled the identification of directional shifts and evolving spatial patterns among the three industrial cities.

The resulting ellipses complement the findings obtained from the spatio-temporal and Kernel Density analyses by providing a generalized representation of the overall direction of industrial development. Accordingly, the directional characteristics derived from the SDE analysis support the development and interpretation of the micro-level spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

4.4 Comparative Spatial Analysis

Following the macro-level and micro-level spatial analyses, a comparative analytical approach was employed to examine differences in industrial specialization among the three industrial cities. Whereas the previous analyses focused on spatial distribution, accessibility, and development patterns, comparative analysis evaluates the relative industrial structure of each city by identifying sectors that exhibit higher or lower levels of specialization. This comparative perspective provides an additional dimension for understanding industrial competitiveness and complements the spatial analyses presented in the previous sections.

4.4.1 Location Quotient (LQ)

Location Quotient (LQ) is one of the most widely used comparative indicators for measuring regional industrial specialization. The method evaluates the relative concentration of a particular industry within a local area by comparing its proportion to that of a reference region. Consequently, LQ provides an effective measure for identifying industries that exhibit comparative specialization or underrepresentation within different geographic areas (Mattos, 2024).

The Location Quotient was calculated using the following equation:

$$LQ = \frac{\left(\frac{e_i}{e}\right)}{\left(\frac{E_i}{E}\right)}$$

Equation 4.2. Location Quotient (LQ) equation used to measure industrial specialization.

where:

- $(e_{\{i,j\}})$ = Number of industrial facilities belonging to industry (i) in industrial city (j).
- (e_j) = Total number of industrial facilities within industrial city (j).
- (E_i) = Total number of industrial facilities belonging to industry (i) across the three industrial cities.
- (E) = Total number of industrial facilities included in the study.

To ensure consistent comparison among the three industrial cities, the analysis was based on a standardized sample consisting of 40 industrial facilities from each industrial city, resulting in a total sample of 120 facilities. Using equal sample sizes minimized potential sampling bias and allowed industrial specialization to be compared directly across the three cities.

The resulting Location Quotient values were interpreted according to conventional thresholds. Values greater than 1 indicate that an industry is more concentrated within an industrial city than in the reference region, suggesting industrial specialization. Values close to 1 represent proportional industrial representation comparable to the regional average, whereas values below 1 indicate that the industry is relatively underrepresented. Following the recommendation of Mattos (2024), values exceeding approximately 1.25 were considered to indicate a relatively strong level of industrial specialization. A threshold of $LQ > 1.25$ was used in this chapter as a stricter descriptive criterion for pronounced specialization, whereas the Industrial Specialization Index (ISI) calculation in Chapter 6 adopts the standard $LQ > 1.00$ criterion to include all sectors exhibiting relative specialization.

The calculated LQ values were used to compare the industrial structures of the three industrial cities and identify their dominant manufacturing sectors. The results support the development and interpretation of the comparative spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

4.5 Industrial Development Readiness Model (IDRM) Framework

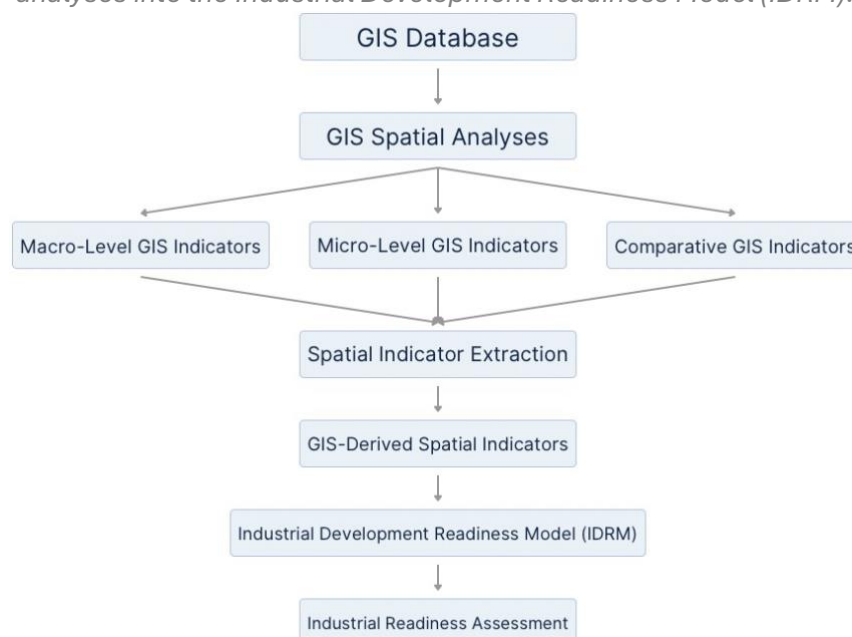
The Industrial Development Readiness Model (IDRM) provides the conceptual framework that links the GIS analyses conducted throughout this study to the industrial readiness assessment. Rather than relying directly on raw GIS outputs, the framework systematically transforms the results of multiple spatial analyses into standardized GIS-derived spatial indicators that collectively represent the different dimensions of industrial development readiness.

The spatial analyses conducted in this study are organized into three complementary analytical levels. Macro-level GIS indicators describe the overall spatial characteristics of industrial cities, including industrial density and logistics infrastructure accessibility. Micro-level GIS indicators capture the spatial behavior and development dynamics of individual industrial facilities through spatio-temporal analysis, Kernel Density Estimation (KDE), and Standard Deviation Ellipse (SDE). Comparative GIS indicators evaluate the relative industrial characteristics of each industrial city using the Location Quotient (LQ) to identify patterns of industrial specialization.

Following the completion of these analyses, the resulting GIS outputs undergo a spatial indicator extraction process, whereby the analytical findings are interpreted and converted into measurable GIS-derived spatial indicators. This process transforms descriptive spatial analysis results into standardized variables suitable for industrial readiness assessment.

The extracted GIS-derived spatial indicators constitute the analytical foundation of the Industrial Development Readiness Model (IDRM). These indicators are subsequently integrated within the model to evaluate and compare the development readiness of the three industrial cities. The procedures for indicator standardization, weighting, and composite index development are presented in Chapter 6.

Figure 4.3. Conceptual framework illustrating the transformation of GIS-derived analyses into the Industrial Development Readiness Model (IDRM).



Chapter 5 — Spatial Analysis Results

5.1 Macro-Level Spatial Analysis

The macro-level spatial analysis evaluates the overall spatial characteristics of the three industrial cities from a regional perspective. Rather than examining the behavior of individual industrial facilities, this level of analysis considers each industrial city as an integrated spatial unit, allowing regional patterns of industrial concentration and accessibility to be investigated. Consequently, the macro-level analysis provides a broad understanding of the spatial structure of industrial development across the study area and establishes the regional context for the subsequent micro-level and comparative analyses.

Two complementary Geographic Information System (GIS) techniques were employed at this analytical level. Industrial Density analysis was conducted to evaluate the overall concentration of industrial facilities within each industrial city, while Buffer Analysis was used to examine the spatial accessibility of industrial facilities to major logistics infrastructure, including road networks, railway stations, airports, ports, and oil and gas infrastructure. Together, these analyses provide macro-level spatial indicators that describe the regional characteristics of industrial development and contribute to the analytical foundation of the Industrial Development Readiness Model (IDRM).

5.1.1 Industrial Density

Industrial Density analysis was conducted to examine the spatial concentration of industrial facilities within the three industrial cities. By calculating the number of industrial facilities relative to the total industrial-city area, the analysis provides a standardized measure of industrial concentration, allowing direct comparison among industrial cities of different spatial extents. This indicator reflects the overall intensity of industrial development and supports the identification of areas exhibiting relatively high or low levels of industrial concentration.

Industrial Density (Factories/km²)

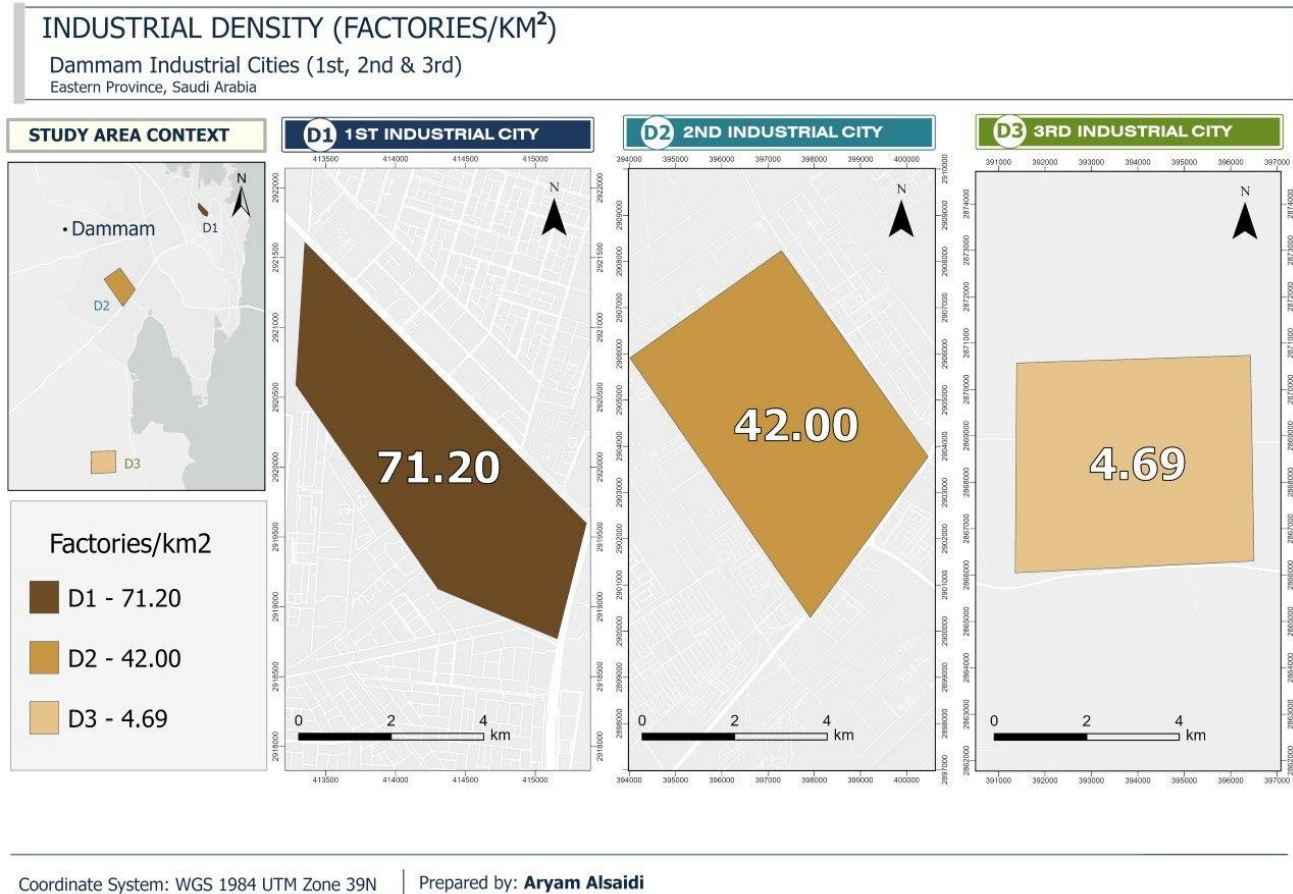


Figure 5.1

Interpretation of Industrial Density Results

The Industrial Density map illustrates clear differences in the concentration of industrial activities among the three industrial cities. The results reveal noticeable spatial variation in the intensity of industrial land utilization across the study area.

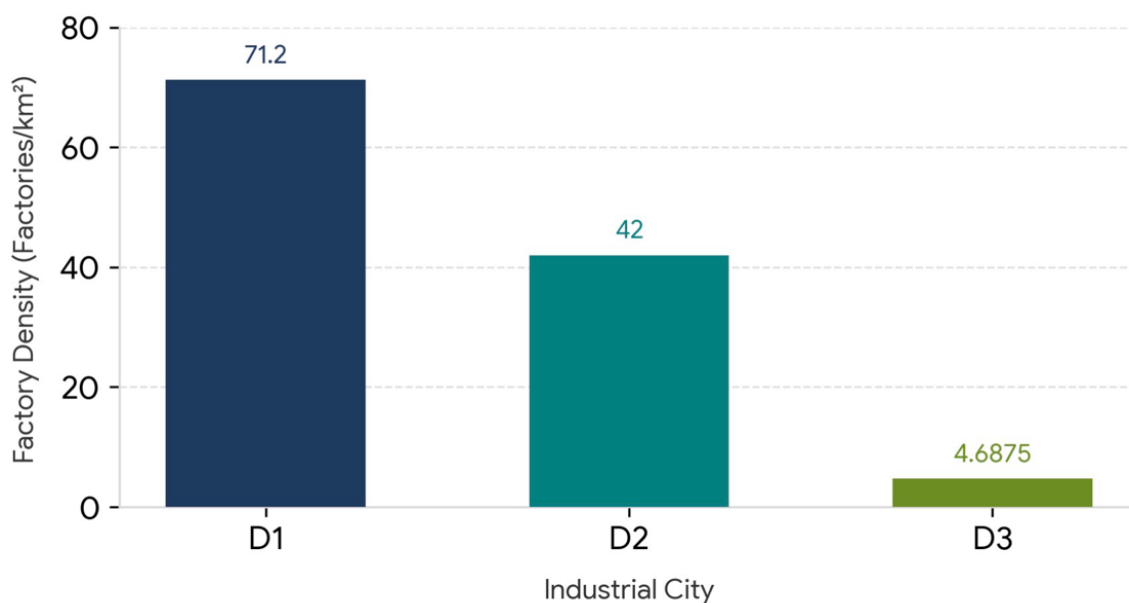
The 1st Industrial City (D1) recorded the highest industrial density (71.20 factories/km²), indicating the greatest concentration of industrial activities within its developed area. The 2nd Industrial City (D2) exhibited a moderate industrial density (42.00 factories/km²), reflecting a more balanced distribution of industrial activities across a substantially larger industrial area. In contrast, the 3rd Industrial City (D3) recorded the lowest industrial density (4.69 factories/km²), reflecting both its extensive land area and its relatively early stage of industrial development, where considerable industrial land remains available for future expansion.

Overall, the spatial distribution demonstrates that industrial density varies considerably among the three industrial cities, indicating that the intensity of industrial development is influenced by both the spatial extent and the development stage of each industrial area.

Table 5.1 Factory Density Statistics

Industrial City	Area (m ²)	Official Factories	Factory Density
D1	2,500,000	178	71.20
D2	25,500,000	1,071	42.00
D3	48,000,000	225	4.69

Figure 5.2 Factory Density Comparison



Statistical Interpretation of Industrial Density

The statistical results are consistent with the spatial patterns illustrated in Figure 5.1. Although the 1st Industrial City (D1) accommodates only 178 factories, it recorded the highest industrial density (71.20 factories/km²) because of its relatively compact developed area. In contrast, the 2nd Industrial City (D2) contains the largest number of factories (1,071), yet exhibits a lower industrial density (42.00 factories/km²), indicating that industrial activities are distributed across a substantially larger geographic area. Meanwhile, the 3rd Industrial City (D3) recorded the lowest industrial density (4.69 factories/km²) despite containing 225 factories, primarily because of its extensive land area and its continuing stage of industrial development.

Overall, the statistical findings indicate that industrial density is influenced not only by the total number of industrial facilities but also by the spatial extent and development stage of each industrial city. Consequently, relatively compact industrial cities tend to exhibit higher density values, whereas larger or developing industrial cities generally accommodate industrial activities over wider areas while maintaining lower overall density levels due to greater land availability.

5.2 Logistics Infrastructure Accessibility Analysis

Logistics infrastructure is one of the primary factors influencing industrial development, as it facilitates the efficient movement of raw materials, finished products, and supporting services across industrial regions. The accessibility of industrial facilities to regional transportation and logistics networks enhances operational efficiency, strengthens supply chain connectivity, and contributes to long-term industrial competitiveness.

To examine these spatial relationships, a series of GIS-based Buffer Analyses was conducted to evaluate the accessibility of industrial facilities to the major logistics infrastructure supporting the study area. The analysis includes five infrastructure categories: major roads, railway infrastructure, airport facilities, port infrastructure, and oil and gas infrastructure. As these logistics components are distributed both within and beyond the administrative boundaries of Dammam, the assessment considers the broader regional logistics network that directly supports the operation and development of the three industrial cities.

The following sections present the accessibility patterns identified for each logistics infrastructure category, highlighting their spatial relationship with industrial facilities and their potential contribution to industrial development readiness.

5.2.1 Road Accessibility

Road accessibility was analyzed to evaluate the spatial relationship between industrial facilities and the major road network within the study area. Efficient road connectivity plays a critical role in supporting industrial activities by facilitating the movement of raw materials, finished products, employees, and logistics services. Consequently, proximity to major roads represents one of the key spatial indicators of industrial accessibility and operational efficiency.

Using the GIS-based Multiple Ring Buffer analysis described in Chapter 4, industrial facilities were evaluated according to their proximity to the major road network within three accessibility zones (5 km, 10 km, and 15 km). This analysis provides a standardized assessment of road accessibility across the three industrial cities and identifies the extent to which industrial facilities benefit from direct access to regional transportation infrastructure.

Accessibility of Industrial Facilities to Major Roads

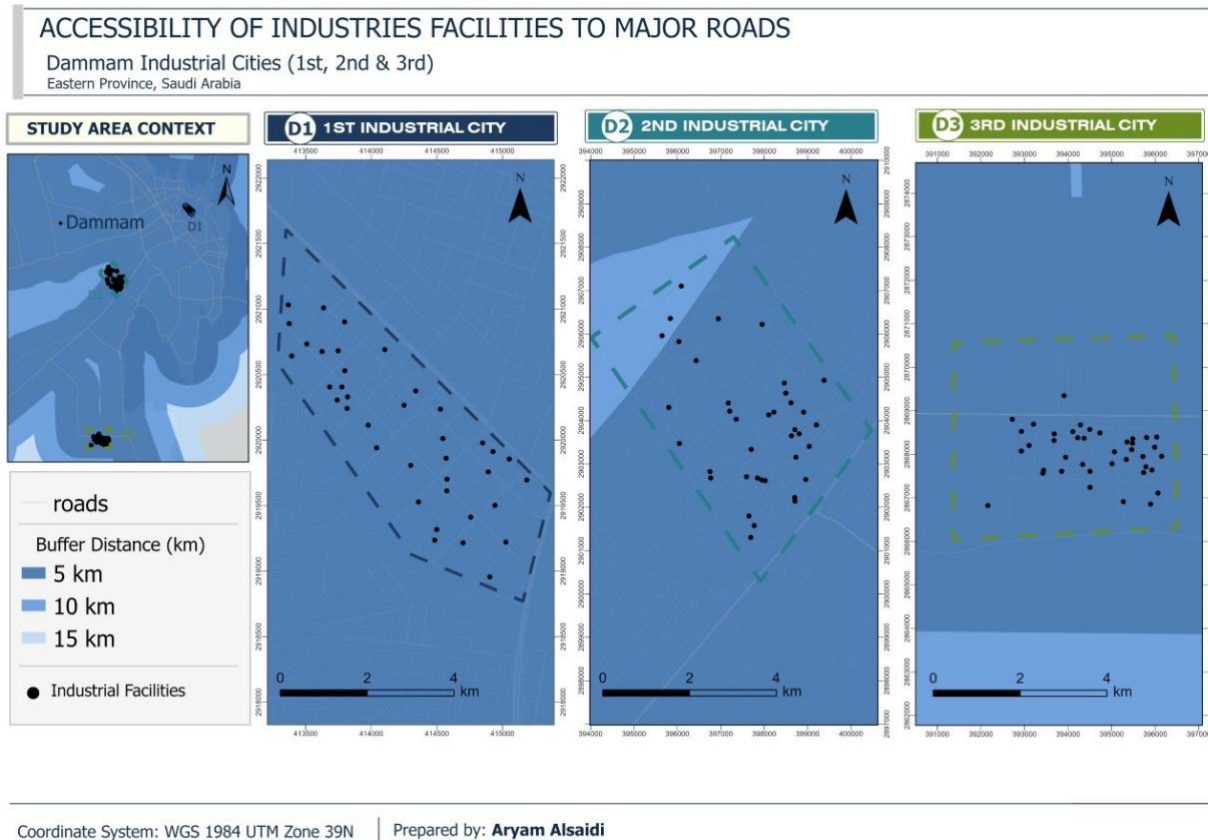


Figure 5.3

Interpretation of Road Accessibility Results

The road accessibility analysis indicates generally high accessibility to the major road network across the three industrial cities, although some spatial variation is observed.

The 1st Industrial City (D1) demonstrates the highest level of accessibility, with all industrial facilities located within the 5 km buffer zone, indicating excellent connectivity to the major road network. Similarly, the 2nd Industrial City (D2) exhibits high accessibility, as most industrial facilities fall within the 5 km buffer, while only a limited number are located within the 10 km buffer zone.

In contrast, the 3rd Industrial City (D3) shows greater variation in accessibility. Most facilities in the northern section are located within the 5 km buffer, whereas facilities in the southern part extend into the 10 km and 15 km buffer zones, reflecting relatively lower accessibility in these areas.

Overall, D1 and D2 benefit from consistently high road accessibility, whereas D3 exhibits less uniform accessibility due to its larger spatial extent. These results provide one of the logistics-related spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

5.2.2 Railway Accessibility

Railway accessibility was analyzed to evaluate the spatial relationship between industrial facilities and the regional railway network. Railway connectivity is particularly important for industrial activities requiring efficient movement of bulk materials and heavy freight, as it provides an alternative transportation mode that supports long-distance logistics and strengthens connections with ports and other industrial regions.

Using the GIS-based Multiple Ring Buffer analysis described in Chapter 4, industrial facilities were evaluated according to their proximity to the railway network within three accessibility zones (5 km, 10 km, and 15 km). This analysis provides a standardized assessment of railway accessibility across the three industrial cities.

Accessibility of Industrial Facilities to Railway

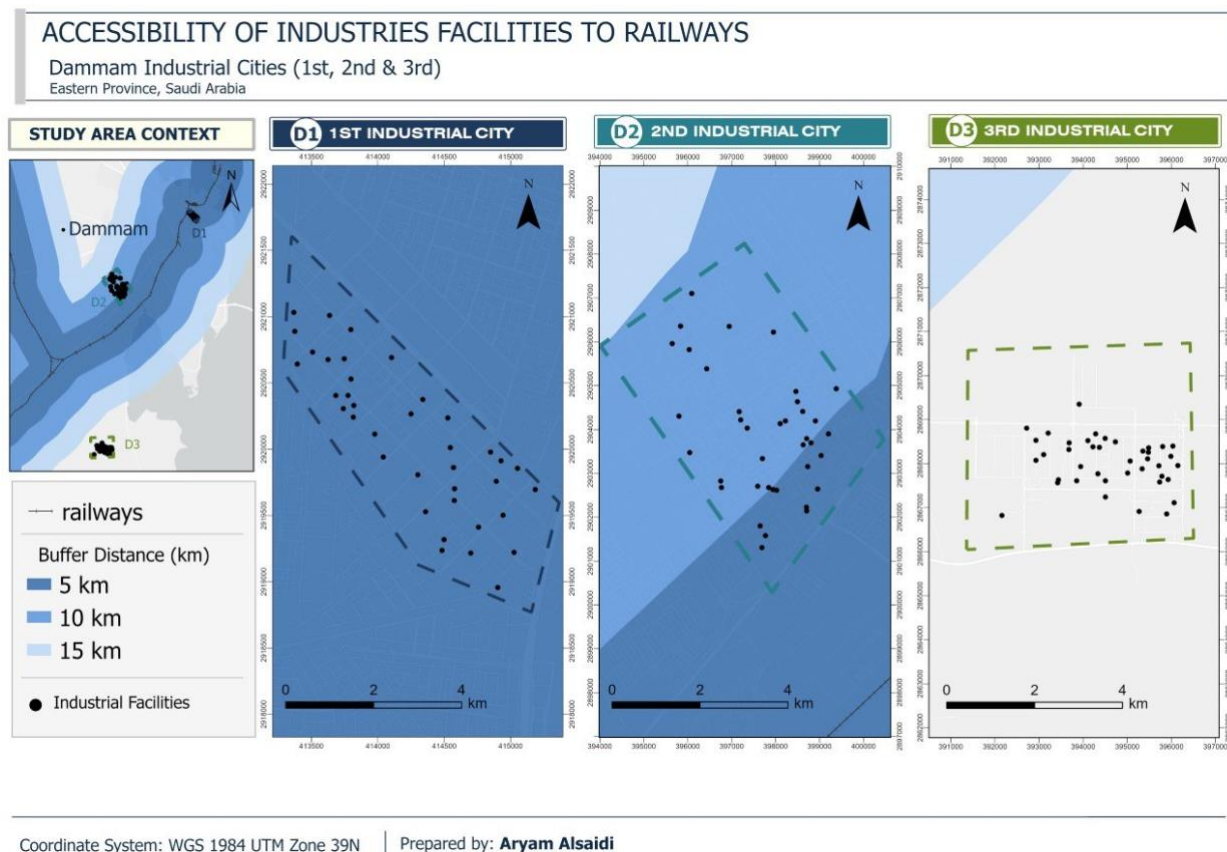


Figure 5.4

Interpretation of Railway Accessibility Results

The railway accessibility analysis reveals considerable spatial variation among the three industrial cities.

The 1st Industrial City (D1) demonstrates the highest level of railway accessibility, with all industrial facilities located within the 5 km buffer zone. This indicates excellent proximity to the railway network, providing favorable conditions for freight transportation and regional logistics connectivity.

The 2nd Industrial City (D2) exhibits moderate railway accessibility. Industrial facilities are distributed across the 5 km, 10 km, and 15 km buffer zones, indicating a gradual decrease in accessibility from the southeastern section toward the northwestern part of the industrial city.

In contrast, the 3rd Industrial City (D3) exhibits the lowest level of railway accessibility. Almost the entire industrial city lies outside the 15 km buffer zone, with only a limited area along its northern edge intersecting the outer buffer. This pattern indicates that D3 currently has minimal direct access to the regional railway network and therefore relies primarily on road transportation for freight movement.

Overall, the results indicate that railway accessibility is highest in D1, moderate in D2, and lowest in D3, highlighting substantial differences in rail-based logistics accessibility among the three industrial cities. These findings provide one of the logistics-related spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

5.2.3 Airport and Railway Station Accessibility

Airport and railway station accessibility was analyzed to evaluate the proximity of industrial facilities to two major transportation terminals that support regional industrial logistics. While airports facilitate the rapid movement of high-value and time-sensitive goods, railway stations provide important connections for freight transportation and regional supply chains. Together, these facilities strengthen industrial connectivity and improve logistics performance.

Using the GIS-based Multiple Ring Buffer analysis described in Chapter 4, industrial facilities were evaluated according to their proximity to airport and railway station facilities within three accessibility zones (5 km, 10 km, and 15 km).

Accessibility of Industrial Facilities to Airport&Rail Station

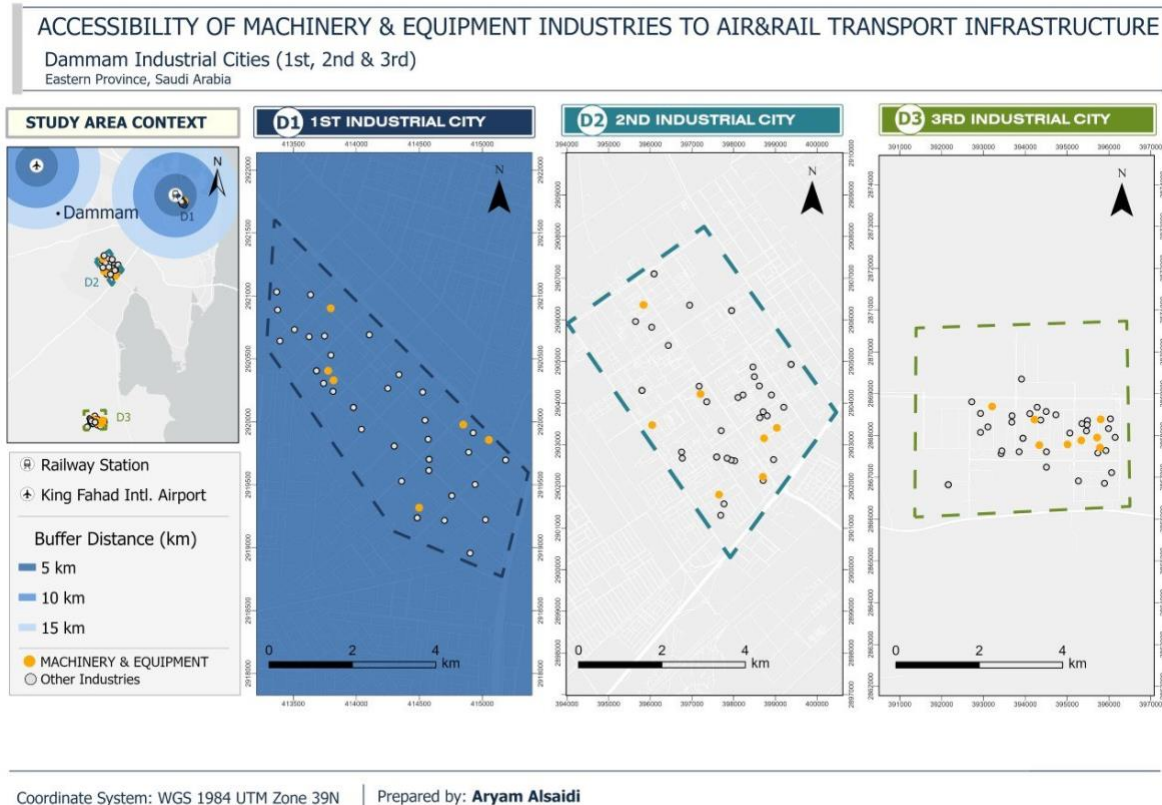


Figure 5.5

Interpretation of Airport&Railway Station Accessibility Results

The accessibility analysis indicates noticeable spatial differences among the three industrial cities in relation to airport and railway station infrastructure.

The 1st Industrial City (D1) demonstrates the highest level of accessibility, with all industrial facilities, including the machinery and equipment industries, located within the 5 km buffer zone. This indicates excellent proximity to both transportation terminals, providing favorable conditions for efficient freight movement and logistics operations.

In contrast, both the 2nd Industrial City (D2) and the 3rd Industrial City (D3) are located outside the 15 km buffer zone, indicating limited direct accessibility to the airport and railway station. However, the spatial context reveals an important distinction between the two cities. Although D2 falls outside the defined buffer distance, it remains relatively close to the transportation terminals, suggesting a moderate level of logistical accessibility through the surrounding road network. Conversely, D3 is located considerably farther from both facilities, reflecting a lower level of accessibility and greater dependence on road transportation for regional and national logistics.

The distribution of machinery and equipment industries further highlights the strategic importance of transportation accessibility. In D1, these industries benefit from immediate proximity to both airport and railway infrastructure, whereas similar industries located in D2 and particularly D3 operate with comparatively lower direct accessibility to these major logistics facilities.

Overall, the results indicate that D1 possesses the strongest air and rail accessibility, D2 occupies an intermediate position despite lying outside the 15 km buffer, and D3 exhibits the lowest level of accessibility because of its greater spatial separation from the transportation terminals.

5.2.4 Port Accessibility

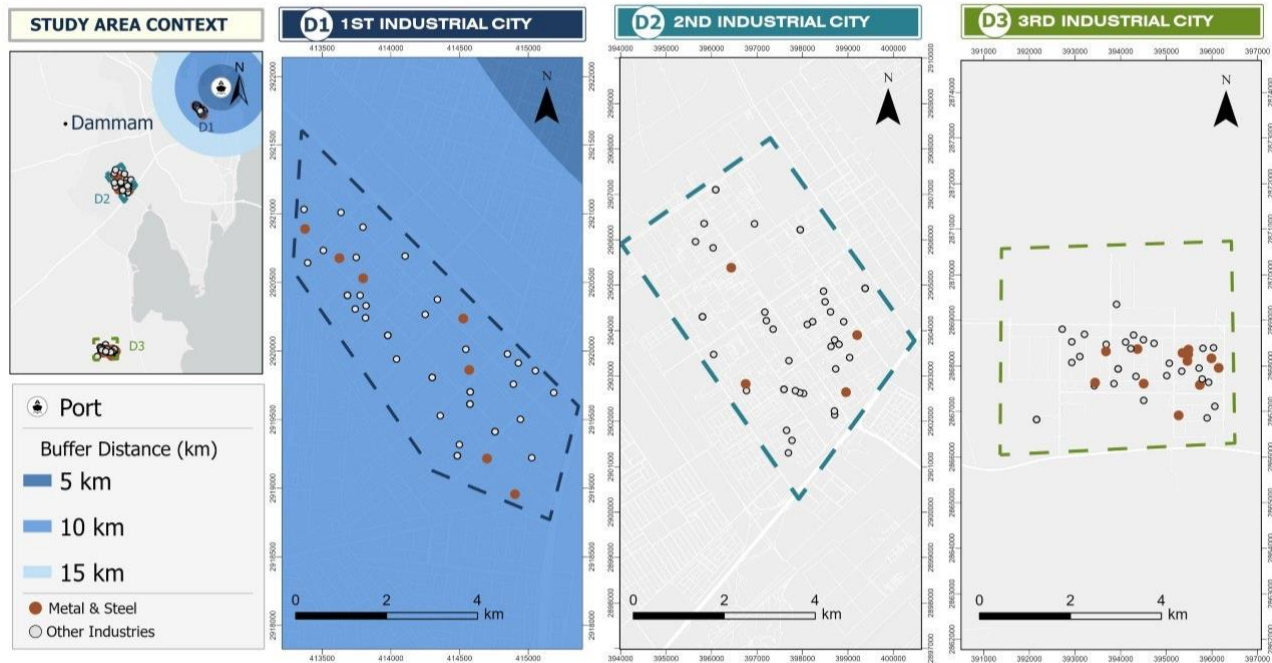
Port accessibility was analyzed to evaluate the proximity of industrial facilities to maritime transportation infrastructure. Ports play a vital role in industrial logistics by facilitating the import of raw materials and the export of manufactured products, particularly for industries that rely on high-volume freight movement. Consequently, accessibility to port infrastructure represents an important indicator of regional industrial connectivity.

Using the GIS-based Multiple Ring Buffer analysis described in Chapter 4, industrial facilities were evaluated according to their proximity to port infrastructure within three accessibility zones (5 km, 10 km, and 15 km). The resulting accessibility patterns provide an additional logistics indicator for assessing industrial development readiness.

Accessibility of Industrial Facilities to port Infrastructure

ACCESSIBILITY OF METAL & STEEL INDUSTRIES TO PORT INFRASTRUCTURE

Dammam Industrial Cities (1st, 2nd & 3rd)
Eastern Province, Saudi Arabia



Coordinate System: WGS 1984 UTM Zone 39N

Prepared by: **Aryam Alsaidi**

Figure 5.6

Interpretation of Port Accessibility Results

The port accessibility analysis reveals substantial differences among the three industrial cities.

The 1st Industrial City (D1) exhibits the highest level of accessibility, with all industrial facilities located within the 5 km buffer zone. This indicates excellent proximity to King Abdulaziz Port, providing favorable conditions for importing raw materials and exporting industrial products through maritime transport.

In contrast, both the 2nd Industrial City (D2) and the 3rd Industrial City (D3) are located outside the 15 km buffer zone, indicating limited direct accessibility to the port. Although D2 is relatively closer to the port than D3, both industrial cities rely primarily on road transportation to connect with maritime logistics facilities.

The spatial distribution of metal and steel industries further illustrates the importance of port accessibility. While D1 benefits from immediate access to port infrastructure, major concentrations of metal and steel industries in D3 are located considerably farther from the port, increasing their dependence on inland freight transportation.

Overall, the analysis indicates that D1 possesses the strongest maritime accessibility, whereas D2 and D3 experience progressively lower levels of direct port accessibility.

5.2.5 Oil and Gas Infrastructure Accessibility

Oil and gas infrastructure accessibility was analyzed to evaluate the proximity of industrial facilities to strategic energy infrastructure. Oil and gas facilities represent a critical component of industrial logistics, particularly for industries that depend on continuous access to hydrocarbon feedstocks and energy resources. Therefore, spatial accessibility to this infrastructure provides an important indicator of industrial operational support and regional competitiveness.

Using the GIS-based Multiple Ring Buffer analysis described in Chapter 4, industrial facilities were evaluated according to their proximity to oil and gas infrastructure within three accessibility zones (5 km, 10 km, and 15 km). The resulting accessibility patterns provide an additional logistics indicator for assessing industrial development readiness.

Accessibility of Industrial Facilities to Oil&Gas Infrastructure

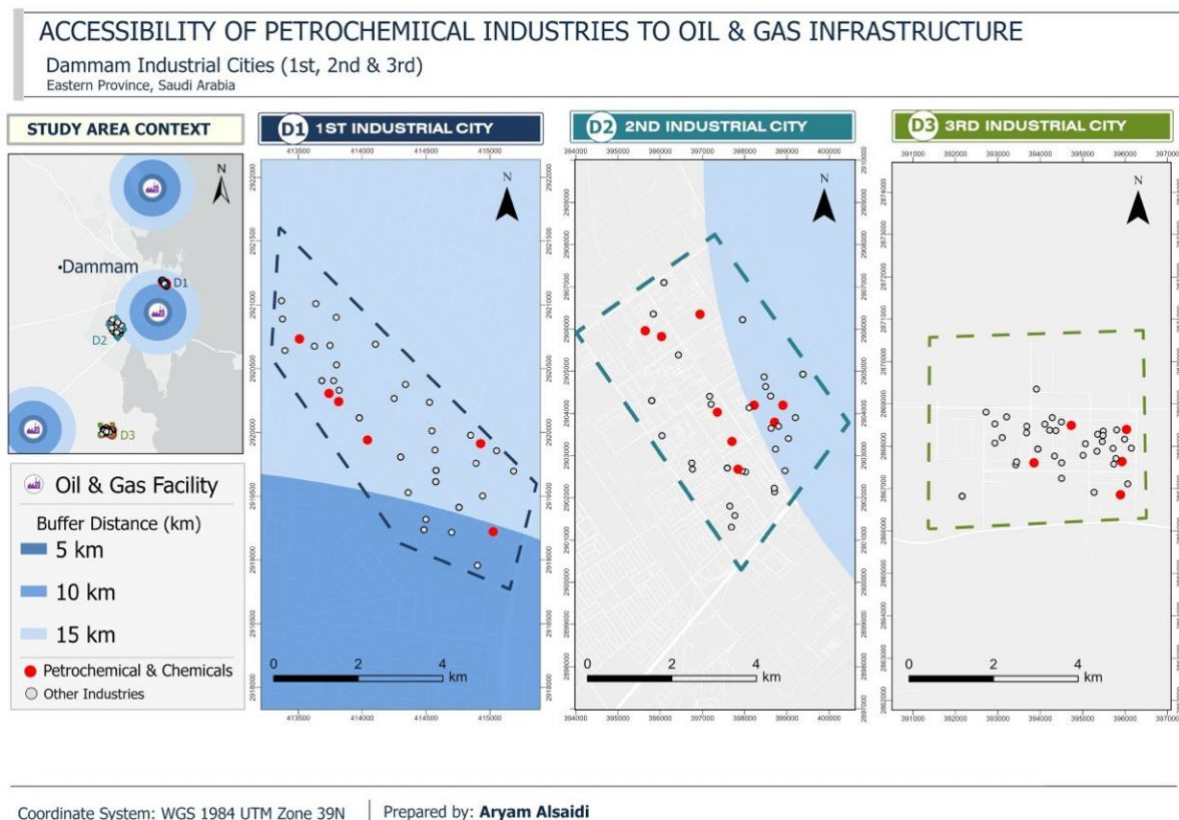


Figure 5.7

Interpretation of Oil&Gas Accessibility Results

The oil and gas infrastructure accessibility analysis reveals different levels of accessibility among the three industrial cities.

The 1st Industrial City (D1) exhibits the highest level of accessibility, with most industrial facilities located within the 10 km buffer zone, while a limited number of facilities fall within the 5 km buffer. This indicates strong proximity to strategic energy infrastructure and favorable conditions for industries requiring continuous access to hydrocarbon resources.

The 2nd Industrial City (D2) demonstrates moderate accessibility. Industrial facilities are primarily distributed across the 10 km and 15 km buffer zones, indicating relatively good accessibility, particularly for facilities located in the eastern part of the industrial city, while western facilities are positioned farther from the energy infrastructure.

In contrast, the 3rd Industrial City (D3) exhibits the lowest level of accessibility, with almost all industrial facilities located outside the 15 km buffer zone. This pattern indicates limited direct access to oil and gas infrastructure and greater dependence on secondary transportation systems for the movement of energy-related resources.

The spatial distribution of petrochemical and chemical industries further emphasizes the importance of energy accessibility. While petrochemical facilities in D1 and parts of D2 benefit from relatively close proximity to oil and gas infrastructure, similar industries located in D3 operate at greater distances from these strategic resources.

Overall, the analysis indicates that D1 has the highest accessibility to oil and gas infrastructure, D2 demonstrates moderate accessibility, and D3 exhibits the lowest level of accessibility. These findings provide the final logistics-related spatial indicator incorporated into the Industrial Development Readiness Model (IDRM).

5.2.6 Summary of Logistics Infrastructure Accessibility

The previous analyses examined the accessibility of industrial facilities to five major logistics infrastructure components, including the road network, railway network, airport and railway station facilities, port infrastructure, and oil and gas infrastructure. Although each analysis examined a different component of logistics infrastructure, collectively they provide a comprehensive overview of the logistics environment supporting industrial development within the three industrial cities. To facilitate comparison, the accessibility patterns identified in the previous sections are summarized and interpreted collectively to highlight the overall logistics strengths and limitations of each industrial city.

Table 5.2. Summary of logistics infrastructure accessibility across the three industrial cities.

Logistics Infrastructure	D1	D2	D3
Road	Within 5 km	Predominantly within 5 km <i>(limited areas within 10 km)</i>	Within 5–15 km
Railway	Within 5 km	Within 5–15 km	Outside 15 km
Airport & Railway Station	Within 5 km	Outside 15 km	Outside 15 km
Port Infrastructure	Within 10 km	Outside 15 km	Outside 15 km
Oil & Gas Infrastructure	Within 10–15 km	Within 10–15 km	Outside 15 km

Figure 5.8. Overall Logistics Infrastructure Accessibility Comparison

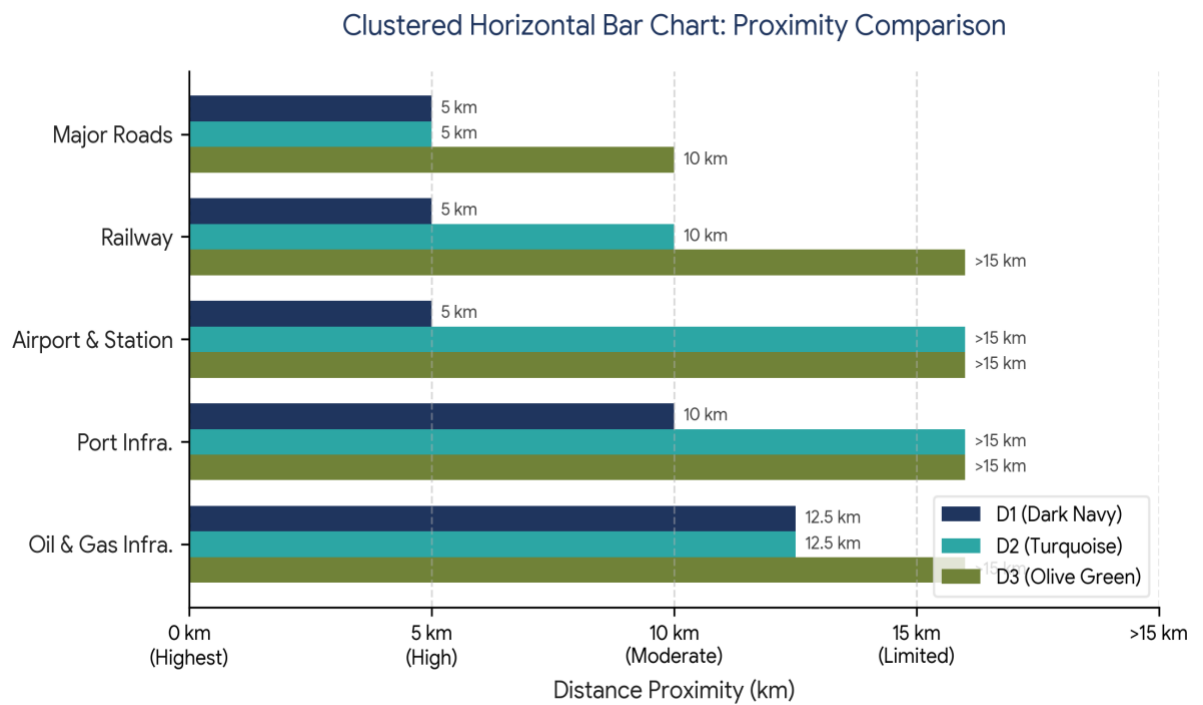
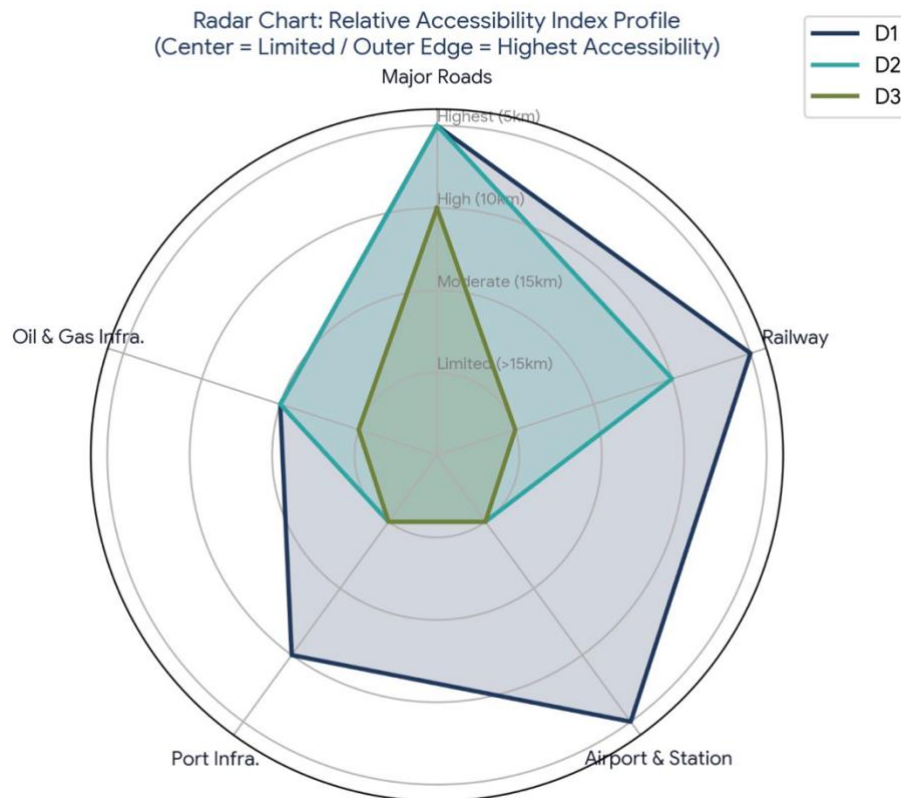


Figure 5.9. Radar comparison of overall logistics infrastructure accessibility.



Overall Interpretation of Logistics Infrastructure Accessibility

The comparative assessment demonstrates clear differences in logistics accessibility among the three industrial cities.

The 1st Industrial City (D1) exhibits the strongest overall logistics accessibility. Industrial facilities are consistently located within the closest accessibility zones for almost all logistics components. Direct accessibility to the major road network, railway infrastructure, airport and railway station facilities, together with close proximity to port and oil and gas infrastructure, provides D1 with the most integrated logistics environment and the highest level of regional connectivity.

The 2nd Industrial City (D2) demonstrates a balanced level of logistics accessibility. Although several logistics components are located outside the predefined 15 km buffer zones, the city remains relatively close to major transportation infrastructure, particularly through its excellent road accessibility and moderate proximity to the railway and oil and gas infrastructure. Consequently, D2 maintains a relatively well-connected logistics environment despite lower direct accessibility than D1.

In contrast, the 3rd Industrial City (D3) exhibits the lowest overall logistics accessibility. With the exception of the major road network, most logistics infrastructure components are located beyond the 15 km accessibility threshold. This greater spatial separation indicates a stronger dependence on road transportation and highlights opportunities for future infrastructure expansion to improve regional logistics connectivity.

Overall, the logistics accessibility assessment indicates a progressive decrease in accessibility from D1 to D3. While D1 benefits from highly integrated logistics infrastructure, D2 maintains moderate regional accessibility, whereas D3 demonstrates the greatest need for future transportation and logistics improvements. These logistics accessibility indicators constitute one of the principal macro-level spatial inputs of the Industrial Development Readiness Model (IDRM) and form part of the integrated spatial indicator framework used for the industrial readiness assessment presented in Chapter 6.

5.3 Micro-Level Spatial Analysis

While the macro-level analysis examined the overall spatial characteristics of the industrial cities as integrated spatial units, micro-level spatial analysis focuses on the spatial behavior and evolution of individual industrial facilities. This level of analysis provides a more detailed understanding of industrial development by examining the distribution, concentration, and directional patterns of factories over time. Consequently, it enables localized spatial characteristics to be identified that may not be apparent through city-level analyses alone.

To investigate these spatial dynamics, three complementary GIS techniques were employed. Spatio-temporal analysis was used to examine the evolution of industrial facilities throughout the study period (2000–2025), providing insight into long-term patterns of industrial expansion. Kernel Density Estimation (KDE) was subsequently applied to identify areas of high industrial concentration and spatial intensity, while Standard Deviational Ellipse (SDE) analysis was conducted to evaluate the dominant direction and geographic dispersion of industrial development.

Together, these analyses provide a comprehensive assessment of industrial development at the facility level and generate the micro-level spatial indicators incorporated into the Industrial Development Readiness Model (IDRM).

5.3.1 Spatio-Temporal Evolution of Industrial Facilities

This section examines the spatio-temporal evolution of industrial facilities across the three industrial cities between 2000–2025. Kernel Density Estimation (KDE) was employed to visualize the spatial concentration, distribution, and evolution of industrial activities over the study period. By comparing the density surfaces generated for each reference year, changes in industrial growth patterns, spatial clustering, and the geographical expansion of industrial development can be systematically identified and interpreted.

Spatio-Temporal Evolution of Industrial Facilities (KDE).



Figure 5.10

Interpretation of Kernel Density Estimation (KDE) Results

The spatio-temporal maps, supported by Kernel Density Estimation (KDE), reveal a clear evolution in the spatial organization of industrial activities across the three industrial cities between 2000 and 2025. By combining the temporal distribution of industrial facilities with density surfaces, the maps illustrate how industrial development expanded, became spatially concentrated, and gradually redistributed throughout the study period.

In 2000, industrial development was strongly concentrated within the First Industrial City (D1), which exhibited the most extensive high-density zones and represented the dominant industrial core of the study area. The Second Industrial City (D2) displayed only localized areas of moderate industrial concentration, whereas no significant

industrial activity was observed within the Third Industrial City (D3), reflecting its undeveloped status during the initial study period.

By 2015, the spatial pattern began to change noticeably. While D1 retained its role as an established industrial center, new density concentrations emerged within D2, indicating the gradual expansion of industrial activities beyond the original industrial core. During the same period, D3 experienced the initial emergence of industrial clustering, representing the early stages of industrial development within the newly established industrial city.

The density patterns observed **in 2020** demonstrate a continued redistribution of industrial activities across the metropolitan industrial system. Industrial clustering became increasingly evident within both D2 and D3, while D1 maintained a relatively stable spatial configuration with limited outward expansion. These patterns suggest that subsequent industrial growth occurred primarily through the development of newer industrial cities rather than continued concentration within the oldest industrial area.

By 2025, the KDE surfaces indicate that industrial activities were well distributed across all three industrial cities, reflecting the progressive maturation of the regional industrial system. Although each industrial city contained comparable levels of industrial representation, the density maps continued to exhibit distinct internal clustering patterns, demonstrating that industrial activities remained spatially heterogeneous rather than uniformly distributed. The persistence of localized high-density clusters suggests that industrial facilities continue to concentrate around preferred development zones within each industrial city.

Overall, the Kernel Density analysis illustrates a clear transition from a predominantly monocentric industrial structure during the early study period toward a more balanced polycentric industrial configuration by 2025. This spatial evolution reflects the gradual decentralization of industrial development from the original industrial core toward the newer industrial cities, highlighting the long-term expansion and spatial diversification of industrial activities across the Dammam metropolitan area.

Table 5.3. Growth of Industrial Facilities across the Three Industrial Cities (2000–2025)

Industrial City	2000	2015	2020	2025
D1	30	36	36	40
D2	19	35	37	40
D3	0	16	26	40

Figure 5.11. Industrial Facility Growth Trends across Three Dammam Industrial Cities

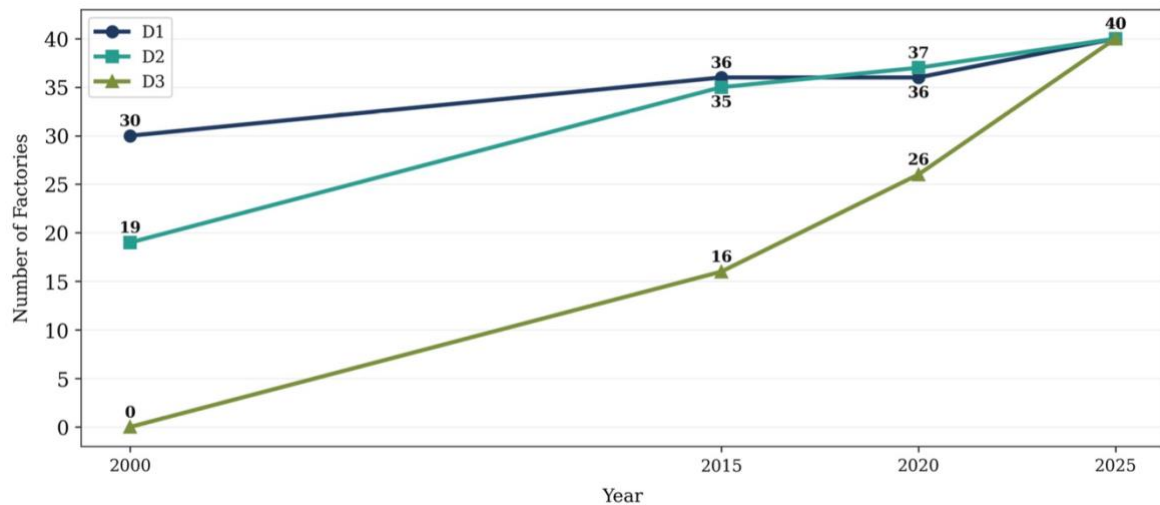


Table 5.1 and Figure 5.2 summarize the temporal growth of industrial facilities across the three industrial cities between 2000 and 2025. While the previous maps illustrate the spatial evolution and density patterns of industrial facilities, the quantitative results presented here describe the numerical growth of industrial facilities across the study period.

The results reveal three distinct development trajectories. Dammam First Industrial City (D1) remained the dominant industrial center during the early years of development, increasing from 30 factories in 2000 to 40 factories in 2025, with no numerical growth recorded between 2015 and 2020, indicating a relatively stable stage of development. In contrast, Dammam Second Industrial City (D2) exhibited continuous expansion throughout the study period, growing from 19 to 40 factories, representing the highest percentage increase among the established industrial cities (110.5%). Dammam Third Industrial City (D3), which contained no industrial facilities in 2000, experienced progressive development after its establishment, increasing from 16 factories in 2015 to 40 factories in 2025, reflecting the emergence of a new industrial center within the metropolitan area.

Overall, the numerical results indicate a gradual transition from an initially concentrated industrial structure toward a more balanced regional distribution. Although the three industrial cities followed different growth trajectories, each city is represented by 40 facilities in 2025 as part of the standardized research sample used in this study. Therefore, the 2025 counts should not be interpreted as equal actual factory totals or equal industrial capacity. Nevertheless, the observed growth patterns demonstrate the progressive expansion of industrial development across the Dammam metropolitan industrial system.

5.3.2 Spatial Direction of Industrial Development (SDE)

The Standard Deviational Ellipse (SDE) analysis was employed to examine the dominant spatial orientation, directional trends, and geographic dispersion of industrial development across the three industrial cities between 2000 and 2025. While the previous analyses illustrated the temporal evolution and concentration of industrial facilities, SDE provides a generalized representation of the overall direction and spatial distribution of industrial growth throughout the study period.

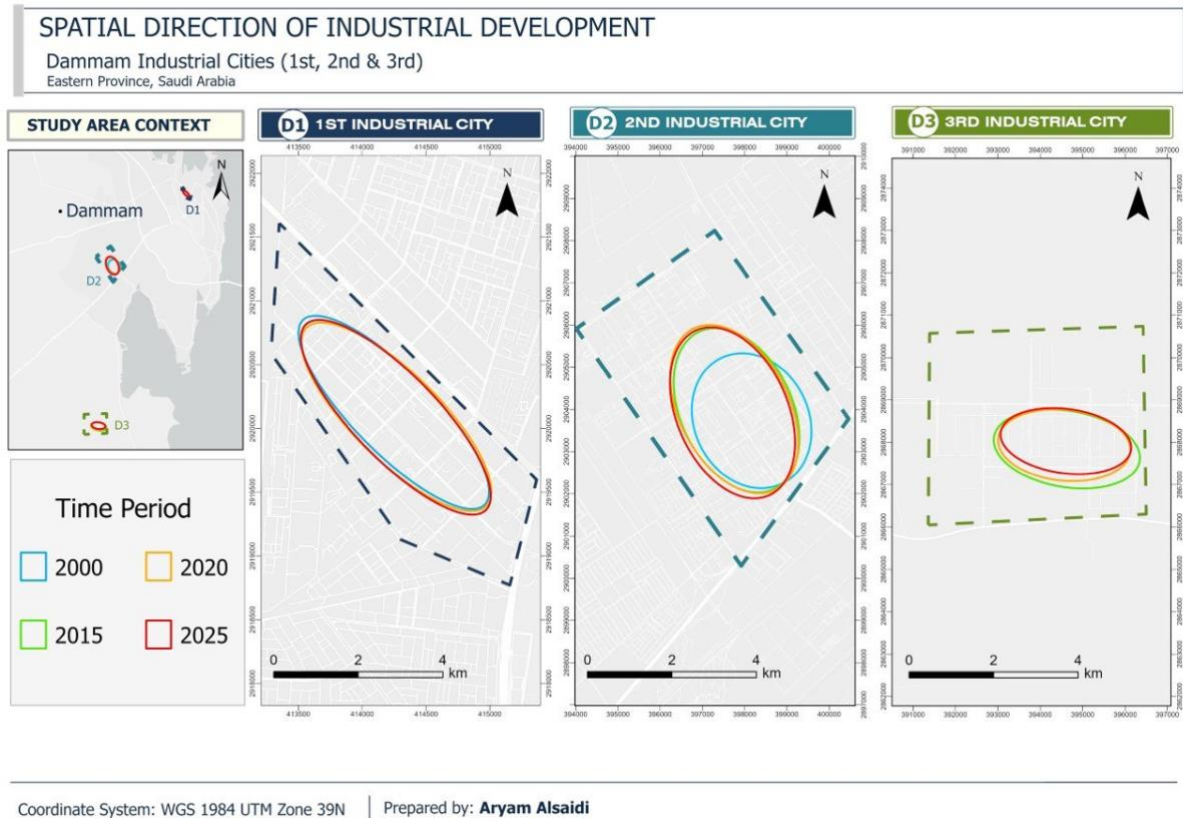


Figure 5.12

Interpretation of Standard Deviational Ellipse (SDE) Results

The Standard Deviational Ellipse (SDE) analysis demonstrates distinct directional characteristics in the spatial evolution of industrial facilities across the three industrial cities between 2000 and 2025. While the previous analyses illustrated the temporal evolution and spatial concentration of industrial facilities, the SDE provides a generalized representation of the dominant spatial orientation, geographic dispersion, and overall directional trend of industrial development throughout the study period. The comparison of ellipses across the four reference years provides a comprehensive representation of how industrial expansion has evolved within each industrial city.

Dammam First Industrial City (D1) exhibits the highest degree of directional stability throughout the study period. The ellipses consistently maintain a northwest–southeast (NW–SE) orientation, indicating that industrial development largely followed an established spatial corridor. Furthermore, the substantial overlap between the 2015

and 2020 ellipses reflects minimal change in the overall spatial distribution during this period, corresponding with the stable number of industrial facilities observed in the quantitative analysis. By 2025, only a slight increase in the spatial extent of the ellipse is observed, while the dominant orientation remains unchanged, indicating limited expansion beyond the existing industrial structure. These characteristics suggest that D1 represents a mature industrial city with a stable spatial development pattern.

In contrast, Dammam Second Industrial City (D2) demonstrates a gradual increase in spatial dispersion throughout the study period. The relatively compact ellipse observed in 2000 progressively expands in subsequent years, indicating that industrial facilities became more spatially distributed as development continued. Although the overall directional orientation remains generally consistent, the increasing size of the ellipse reflects the continuous expansion of industrial activities across a wider geographic area, consistent with the substantial increase in industrial facilities recorded between 2000 and 2025. This pattern reflects the continuous spatial growth of industrial development within D2 while maintaining a relatively stable directional trend.

Dammam Third Industrial City (D3) displays a spatial development pattern distinct from the other two industrial cities. As no industrial facilities were present in 2000, the first Standard Deviational Ellipse appears in 2015 following the establishment of industrial activities within the city. The ellipses consistently exhibit a predominantly east–west (E–W) orientation throughout the remaining study period. While the directional trend remains relatively stable, the gradual enlargement of the ellipse between 2015 and 2025 reflects the continued spatial expansion of industrial development as the city evolved into an important industrial center. This directional pattern is consistent with the relatively recent stage of industrial development observed within D3.

Overall, the Standard Deviational Ellipse analysis reveals three contrasting spatial development patterns within the Dammam metropolitan industrial system. D1 is characterized by long-term directional stability, indicating a mature industrial structure; D2 demonstrates progressive spatial expansion through increasing dispersion, reflecting continuous industrial growth; whereas D3 exhibits a distinct east–west directional growth pattern associated with its ongoing industrial development. These findings complement the spatio-temporal and density analyses by illustrating not only where industrial development became concentrated, but also the dominant directions through which industrial expansion evolved over the 25-year study period.

5.4 Comparative Spatial Analysis

While the previous sections examined the spatial characteristics of industrial development at both the macro- and micro-levels, this section focuses on comparing the industrial structure of the three industrial cities. Rather than evaluating spatial distribution or accessibility alone, the comparative analysis identifies the dominant industrial specializations within each industrial city using the **Location Quotient (LQ)**. *This approach highlights the relative concentration of different industrial sectors and reveals the functional role of each industrial city within the regional industrial system. The results provide an additional comparative indicator that complements the previous spatial analyses and contributes to the overall assessment of industrial development readiness.*

5.4 Comparative Analysis of Industrial Specialization Using Location Quotient (LQ)

Industrial cities differ not only in their spatial distribution and accessibility but also in the composition of their industrial activities. While previous analyses examined the spatial characteristics and logistics environment of the three industrial cities, the present analysis investigates their industrial specialization using the Location Quotient (LQ). This technique identifies industries that are proportionally more concentrated within a particular industrial city compared with the overall study area, thereby revealing the relative industrial strengths and specialization patterns of each city.

The analysis was performed using the industrial database developed for this research, which classifies the sampled industrial facilities into nine industrial sectors. The analysis used a standardized sample of 40 industrial facilities from each industrial city, resulting in a total sample of 120 facilities across the three industrial cities. The calculated LQ values were subsequently used to classify the dominant industrial specialization of each industrial city and were visualized through thematic mapping.

Industrial Specialization of the Three Industrial Cities Based on Location Quotient (LQ)

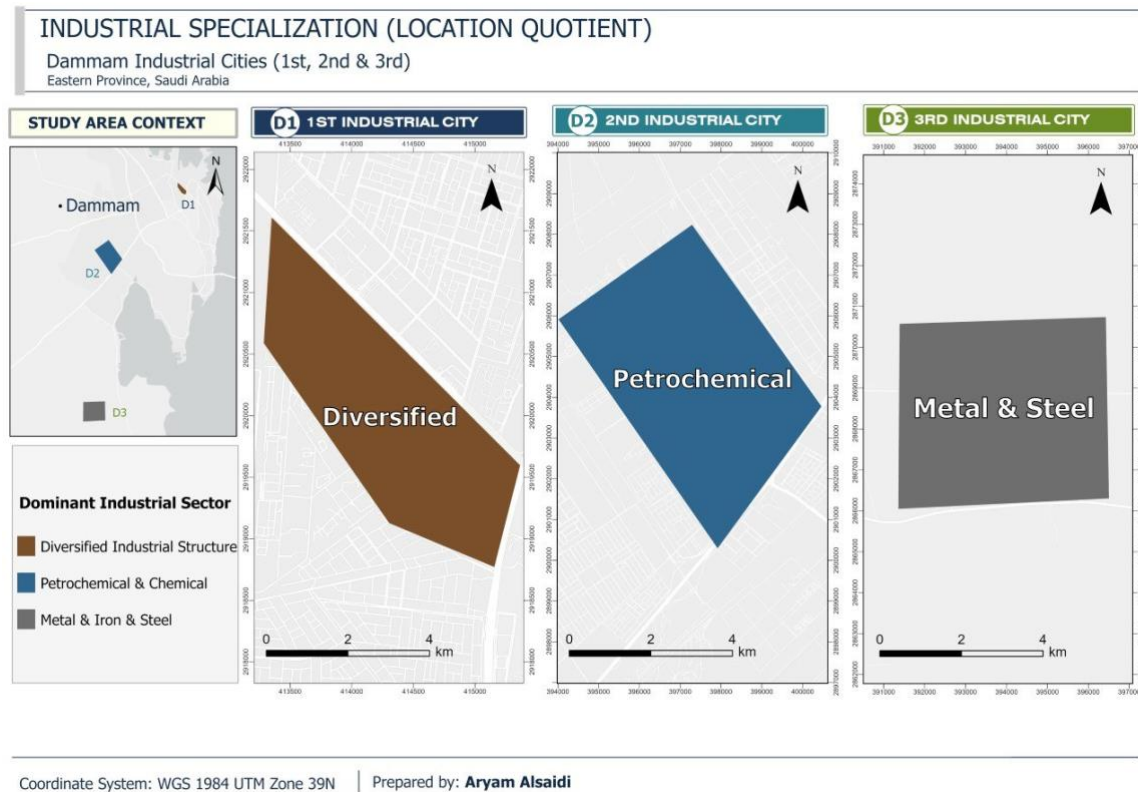


Figure 5.13

Interpretation of Industrial Specialization

The Location Quotient analysis reveals distinct industrial specialization patterns among the three industrial cities, indicating that each city has developed a unique industrial profile despite sharing the same regional industrial system.

The 1st Industrial City (D1) exhibits a diversified industrial structure. Several industrial sectors recorded LQ values greater than 1.0, including Building Materials, Textiles & Paper, Pharmaceutical & Medical Products, and Food & Beverage industries. The relatively balanced distribution of specialization across multiple sectors indicates that D1 functions as a diversified industrial center rather than being dominated by a single industrial activity.

The 2nd Industrial City (D2) demonstrates a clear specialization in Petrochemical & Chemical Industries, which achieved the highest concentration within the city (LQ = 1.35). Plastics & Rubber industries also recorded relatively high specialization values, reflecting the complementary relationship between petrochemical production and downstream manufacturing activities. This pattern identifies D2 as the primary petrochemical manufacturing cluster within the study area.

The 3rd Industrial City (D3) is characterized by a strong specialization in Metal, Iron & Steel Industries, which recorded the highest Location Quotient among the major industrial sectors (LQ = 1.50). In addition, the “Other Manufacturing” category achieved the highest overall LQ value (2.40), although it represents a relatively small number of factories. The concentration of heavy industries indicates that D3 has evolved as the principal heavy industrial zone within the regional industrial system.

Overall, the comparative analysis demonstrates that industrial specialization varies considerably among the three industrial cities. While D1 maintains a diversified industrial economy, D2 specializes primarily in petrochemical industries, whereas D3 exhibits a clear concentration of heavy metal industries. These specialization patterns represent an important comparative spatial indicator and contribute directly to the Industrial Development Readiness Model (IDRM).

Table 5.4. Location Quotient (LQ) Values and Industrial Factory Distribution across the Three Industrial cities (D1, D2, D3)

Industrial Sector	Regional Total (E _i)	D1 Count (e _{i,1})	D1 LQ	D2 Count (e _{i,2})	D2 LQ	D3 Count (e _{i,3})	D3 LQ
Building Materials	14	7	1.50	5	1.07	2	0.43
Food & Beverage	10	4	1.20	4	1.20	2	0.60
Machinery & Equipment	21	6	0.86	7	1.00	8	1.14
Metal / Iron & Steel	24	7	0.88	5	0.63	12	1.50
Other Manufacturing	5	0	0.00	1	0.60	4	2.40
Petrochemical / Chemicals	20	6	0.90	9	1.35	5	0.75
Pharmaceutical / Medical	4	2	1.50	1	0.75	1	0.75
Plastics & Rubber	12	3	0.75	5	1.25	4	1.00
Textiles / Paper	10	5	1.50	3	0.90	2	0.60
Total	120	40	1.00	40	1.00	40	1.00

Note: Location Quotient (LQ) values greater than 1.00 indicate industrial specialization within the corresponding industrial city. Values highlighted in green represent sectors with regional specialization.

Figure 5.14. Location Quotient (LQ) Spatial Specialization Heatmap



5.5 Synthesis of Spatial Findings

The integrated spatial analyses presented in this chapter demonstrate that industrial development characteristics are influenced by the combined effects of industrial concentration, logistics accessibility, spatio-temporal evolution, spatial orientation, and industrial specialization. Rather than functioning as isolated dimensions, these spatial characteristics collectively reveal the relative strengths and limitations of each industrial city and provide the empirical foundation for the Industrial Development Readiness Model (IDRM).

The First Industrial City (D1) demonstrates the strongest current industrial position among the three cities. It combines the highest industrial density with the strongest proximity to major logistics infrastructure, including roads, railways, airport and railway station facilities, port infrastructure, and oil and gas infrastructure. In addition, its diversified industrial structure indicates a mature industrial environment capable of supporting a wide range of manufacturing activities. However, its spatial capacity for future expansion is minimal, and — as demonstrated by the full IDRM assessment in Chapter 6 — its overall development readiness score is lower than that of D3 once expansion potential, industrial growth, and specialization are also taken into account. D1's strategic location nonetheless provides significant present-day operational advantages.

The Second Industrial City (D2) represents a well-balanced industrial environment. While its proximity to logistics infrastructure is lower than that of D1 for several components, it maintains excellent accessibility to the regional road network and moderate accessibility to railway and energy infrastructure. Together with its strong specialization in petrochemical and chemical industries, these characteristics position D2 as a major industrial processing center with considerable potential for continued industrial growth.

The Third Industrial City (D3) exhibits a contrasting spatial profile. The spatio-temporal analyses indicate the most significant industrial expansion during the study period, together with a clear specialization in heavy industries, particularly metal, iron, and steel manufacturing. However, the logistics accessibility assessment shows that most major logistics infrastructure components are located beyond the principal accessibility buffers. Consequently, industrial activities within D3 depend primarily on road accessibility. D3's combination of low current density, low current accessibility, substantial undeveloped land, the strongest absolute growth within the research sample, and strong sectoral specialization is precisely what drives its position as the strategic industrial expansion zone with the highest overall development readiness, formally established in Chapter 6.

Overall, the combined findings demonstrate that industrial development readiness is determined not by a single spatial indicator but by the interaction between industrial concentration, accessibility to logistics infrastructure, spatial growth patterns, and industrial specialization. The analyses further reveal a progressive transition from an

established industrial core (D1) toward newer industrial development areas (D2 and D3), while simultaneously highlighting the importance of logistics accessibility in supporting future industrial expansion.

These integrated spatial findings provide the empirical foundation for the Industrial Development Readiness Model (IDRM) developed in the following chapter, where the spatial indicators are combined to evaluate the overall readiness of each industrial city and support strategic planning recommendations.

Table 5.5. Comparative Summary of the Three Industrial Cities Based on Spatial Analysis Results

Indicator	D1	D2	D3
Total Area (million m ²)	2.50	25.50	48.00
Undeveloped Land (million m ²)	0.00	≈ 0.00	38.00
Official Factories	178	1,071	225
Factory Density (Factories/km ²)	71.20	42.00	4.69
Dominant Industrial Sector	Diversified Industrial Structure	Petrochemical & Chemical	Metal & Iron & Steel
Logistics Accessibility	High	Moderate	Low
Spatial Growth Pattern	Stable	Continuous Expansion	Emerging Expansion
Overall Spatial Role	Mature Industrial Core	Industrial Processing Hub	Strategic Industrial Expansion Zone

Table 5.5 provides an integrated summary of the principal findings obtained from the macro-level, micro-level, and comparative spatial analyses presented in this chapter. It consolidates the key spatial characteristics of each industrial city by combining land availability, industrial density, logistics accessibility, spatio-temporal development patterns, and industrial specialization into a single comparative framework. This synthesis highlights the complementary roles of the three industrial cities and provides a concise overview of their relative industrial development readiness. This integrated summary provides the empirical bridge between the spatial analyses presented in this chapter and the Industrial Development Readiness Model (IDRM) discussed in Chapter 6.

Chapter 6 — Industrial Development Readiness Model (IDRM)

6.1 Framework Overview

Chapter 5 presented the spatial characteristics of the three industrial cities through a combination of macro-level, micro-level, and comparative GIS analyses. These analyses examined industrial density, logistics accessibility, spatio-temporal development, spatial concentration and direction, and industrial specialization. The results demonstrated substantial differences among the First Industrial City (D1), Second Industrial City (D2), and Third Industrial City (D3), indicating that no single spatial characteristic is sufficient to describe the overall development readiness of an industrial city.

The Industrial Development Readiness Model (IDRM) was therefore developed to integrate the principal spatial findings into a single comparative assessment framework. The model converts selected GIS results into five standardized spatial indicators representing complementary dimensions of industrial development: Industrial Density Index (IDI), Logistics Accessibility Index (LAI), Industrial Expansion Potential Index (IEPI), Industrial Growth Index (IGI), and Industrial Specialization Index (ISI).

The model combines findings from the two analytical databases established in Chapters 3 and 4. The official MODON Industrial Cities database supports the IDI, the LAI, and macro-level city characteristics such as undeveloped land used in the IEPI. The author-compiled Industrial Facilities database — 40 factories per city, 120 factories in total — supports the IGI and ISI, together with the micro-level temporal and spatial analyses presented in Chapter 5. The LAI is derived from the logistics accessibility analysis conducted using the spatial relationships between industrial facilities and infrastructure described in Section 5.2.

This structure maintains a direct relationship between the GIS analyses and the final readiness assessment. The IDRM is therefore not based on additional conceptual indicators or external qualitative scores, but on measurable spatial and temporal characteristics derived from the study database and the analyses presented in Chapter 5. The resulting model provides a comparative rather than absolute measure of industrial development readiness. A higher IDRM score indicates a stronger overall position relative to the other industrial cities included in the study.

6.2 Development of Spatial Indicators

Five spatial indicators were developed from the empirical results presented in Chapter 5. Each indicator was designed to represent a distinct dimension of industrial development readiness while maintaining a clear relationship with the original GIS data and with the two-database structure established in Chapters 3 and 4.

6.2.1 Industrial Density Index (IDI)

The Industrial Density Index represents the intensity of industrial activity within each industrial city, based exclusively on official MODON data. Industrial density was calculated by relating the official number of factories to the corresponding total industrial-city area:

$$Di = Ni / Ai$$

where D_i is the industrial density of city i (factories/km²), N_i is the official number of factories in city i , and A_i is the total industrial-city area of city i (km²). The Chapter 5 results recorded densities of 71.20 factories/km² for D1, 42.00 factories/km² for D2, and 4.69 factories/km² for D3. D1 therefore represents the highest level of current industrial concentration, whereas D3 has the lowest density because of its substantially larger spatial extent and continuing development stage. The 120-factory author-compiled sample is not used at any point in this calculation.

The density values were subsequently standardized to a common 1-5 scale in Section 6.3.

6.2.2 Logistics Accessibility Index (LAI)

The Logistics Accessibility Index (LAI) represents the relative accessibility of sampled industrial facilities to major logistics and strategic infrastructure. The index was derived from the multiple ring buffer analyses presented in Chapter 5.

Five infrastructure categories were considered:

- Major roads;
- Railway infrastructure;
- Airport and railway station facilities;
- Port infrastructure; and
- Oil and gas infrastructure.

For each infrastructure category, industrial facilities were classified into three accessibility zones: within 5 km, 5-10 km, and 10-15 km. Facilities located beyond 15 km were considered outside the defined accessibility range.

A weighted accessibility score was assigned to each zone as follows:

- 3 points for facilities within 5 km;
- 2 points for facilities between 5 and 10 km;
- 1 point for facilities between 10 and 15 km; and
- 0 points for facilities beyond 15 km.

For each infrastructure category, the accessibility score was calculated from the proportion of sampled facilities located within each accessibility zone:

$$A_{i,j} = 3 \cdot P(\leq 5) + 2 \cdot P(5-10) + P(10-15)$$

where $A_{i,j}$ is the accessibility score of city i to infrastructure category j ; $P(\leq 5)$ is the proportion of sampled facilities located within 5 km; $P(5-10)$ is the proportion located between 5 and 10 km; and $P(10-15)$ is the proportion located between 10 and 15 km.

The composite Logistics Accessibility Index for each industrial city was then calculated as the arithmetic mean of the five infrastructure-specific accessibility scores:

$$LAI_i = (\sum_j A_{i,j}) / 5$$

The spatial classifications reported in Table 5.2 provide the GIS-based spatial evidence for this calculation, while the numerical proportions were obtained from the sampled industrial-facility dataset used in the facility-level accessibility analysis.

Accordingly, LAI should be interpreted as a comparative accessibility measure based on the sampled industrial facilities rather than as an accessibility measure calculated from the official total number of factories reported at the industrial-city level.

6.2.3 Industrial Expansion Potential Index (IEPI)

The Industrial Expansion Potential Index represents the remaining spatial capacity available for future industrial development, based on official MODON land-use data.

The indicator was derived from the relationship between undeveloped industrial land and total industrial-city area:

$$E_i = U_i / A_i$$

where E_i is the expansion potential ratio, U_i is the undeveloped land area, and A_i is the total industrial-city area. The Chapter 5 results indicate that D1 has no reported undeveloped land, while D2 has approximately zero undeveloped land. In contrast, D3 contains approximately 38 million m^2 of undeveloped land within a total industrial-city area of approximately 48 million m^2 .

Consequently, D3 demonstrates substantially greater spatial expansion potential than D1 and D2. The indicator was standardized so that a higher value represents greater potential for future industrial expansion.

KDE and SDE results were used as supporting spatial evidence for interpreting industrial expansion patterns but were not converted into additional numerical components of IEPI because no separate quantitative KDE/SDE expansion metric was established. This maintains methodological transparency and avoids introducing subjective measurements into the index.

6.2.4 Industrial Growth Index (IGI)

The Industrial Growth Index represents the change in industrial activity over the 2000-2025 study period, based exclusively on the author-compiled 40-factory-per-city sample.

The temporal database contains factory-status information for 2000, 2015, 2020, and 2025, enabling the development of a reproducible temporal growth measure.

Because D3 recorded zero sampled factories in 2000, a conventional percentage-growth calculation would result in division by zero. Therefore, the model uses the absolute increase in the number of sampled factories between 2000 and 2025:

$$Gi = Ni,2025 - Ni,2000$$

Table 6.1. Absolute Industrial Growth by City, 2000-2025 (Author-Compiled Sample)

Industrial City	2000	2025	Absolute Increase
D1	30	40	10
D2	19	40	21
D3	0	40	40

The results demonstrate the strongest absolute industrial expansion in D3, followed by D2 and D1. This quantitative result is consistent with the temporal analysis presented in Chapter 5, where D3 was identified as the city exhibiting the most significant industrial expansion during the study period.

The absolute growth values were subsequently standardized to the common 1-5 scale.

6.2.5 Industrial Specialization Index (ISI)

The Industrial Specialization Index (ISI) represents the relative degree and breadth of industrial specialization within each industrial city. The index was derived from the Location Quotient (LQ) analysis presented in Chapter 5 using the standardized sample of 40 factories per city, resulting in a total sample of 120 factories across nine industrial sectors.

An LQ value greater than 1.00 indicates that a sector is relatively more concentrated within a city than in the overall regional sample. For descriptive interpretation, an LQ greater than 1.25 may be used to identify relatively pronounced specialization; however, the ISI calculation adopts the standard $LQ > 1.00$ criterion so that all relatively specialized sectors are included.

Rather than relying only on the single highest LQ value, the ISI incorporates the strength of specialization across all sectors satisfying the $LQ > 1.00$ criterion:

$$ISI_i = [\sum (s \in S_i) (LQ_{i,s} - 1)] / |S_i|$$

where S_i is the set of industrial sectors with $LQ_{i,s} > 1.00$ in city i , and $LQ_{i,s}$ is the Location Quotient of sector s in city i .

Based on the LQ values presented in Table 5.4, the resulting raw ISI values are 0.425 for D1, 0.2175 for D2, and 0.680 for D3. These values indicate that D3 exhibits the strongest

average sectoral specialization within the sampled industrial structure, followed by D1 and D2.

The raw ISI values were subsequently standardized to the common 1-5 scale using the min-max normalization procedure described in Section 6.3.

6.3 Indicator Standardization

The five indicators are measured using different units and scales. Industrial density is expressed as factories/km², logistics accessibility is derived from spatial buffer proportions, expansion potential is represented by a land-availability ratio, industrial growth is expressed as a change in factory count, and industrial specialization is derived from LQ values.

To enable their integration into one composite index, all five indicators were transformed into a common 1-5 scale using min-max normalization. For indicators in which higher values represent stronger readiness, the standardized indicator was calculated as:

$$li = 1 + 4 \times [(Xi - Xmin) / (Xmax - Xmin)]$$

where li is the standardized indicator value for city i , Xi is the raw indicator value, and $Xmin$ and $Xmax$ are the minimum and maximum raw values among the three cities. This transformation preserves the relative ordering of the cities while placing all indicators within the same 1-5 range. The normalization is relative to the three cities examined in this study; a standardized score of 5 indicates the strongest performer among D1, D2, and D3 for that indicator, not an absolute real-world benchmark.

Table 6.2. Standardized Spatial Indicators (1-5 Scale)

Indicator	D1	D2	D3
IDI	5.00	3.24	1.00
LAI	5.00	1.99	1.00
IEPI	1.00	1.00	5.00
IGI	1.00	2.47	5.00
ISI	2.79	1.00	5.00

The standardized values demonstrate the contrasting development profiles of the three industrial cities. D1 performs strongly in current industrial density and logistics accessibility, while D3 performs strongly in expansion potential, industrial growth, and specialization. D2 generally occupies an intermediate position.

6.4 Weighting Procedure

Following standardization, an equal-weighting approach was adopted for the five indicators.

Equal weighting was selected because the present study did not employ a formal expert elicitation process, the Analytic Hierarchy Process (AHP), entropy weighting, or any other statistical weighting procedure capable of empirically establishing different levels of importance among the five indicators.

Given the exploratory and proof-of-concept nature of the proposed model, equal weighting was considered the most transparent and reproducible weighting strategy.

Accordingly, each indicator was assigned an equal weight of 0.20:

$$WIDi = WLAi = WIEPi = WIGi = WISI = 0.20$$

The total weight is therefore:

$$\sum W_j = 5 \times 0.20 = 1.00$$

Equal weighting ensures that no individual indicator dominates the composite model solely because of an unsupported subjective weighting assumption. It also provides a transparent and reproducible baseline for comparing the three industrial cities.

The weighting approach should therefore be understood as a methodological baseline rather than a claim that all dimensions have identical real-world importance. Future applications involving expert elicitation or formal MCDA weighting techniques could test alternative weighting structures.

6.5 Model Formulation

The Industrial Development Readiness Model integrates the five standardized indicators into a single composite readiness score. The final model is expressed as:

$$IDRM_i = 0.20 \times IDi + 0.20 \times LAi + 0.20 \times IEPi + 0.20 \times IGi + 0.20 \times ISi$$

or equivalently:

$$IDRM_i = 0.20 \times (IDi + LAi + IEPi + IGi + ISi)$$

where $IDRM_i$ represents the overall industrial development readiness score for industrial city i . Because each standardized indicator ranges from 1 to 5 and the weights sum to 1.00, the theoretical range of the composite IDRM score is also 1-5.

Table 6.3. Dimensions Represented by the Five IDRM Indicators

Indicator	Dimension Represented	Higher Score Indicates
IDI	Current industrial intensity	Greater industrial concentration
LAI	Logistics connectivity	Better spatial accessibility
IEPI	Future spatial capacity	Greater expansion potential
IGI	Temporal development	Stronger industrial growth
ISI	Industrial structure	Stronger sectoral specialization

The model is consequently designed to distinguish between current operational strength and future development potential rather than treating industrial readiness as a single-dimensional concept.

6.6 Industrial Development Readiness Assessment

The final IDRM scores were calculated using the standardized indicator values and equal weights.

D1 — First Industrial City

$$IDRM(D1) = 0.20 \times (5.00 + 5.00 + 1.00 + 1.00 + 2.79) = 2.96$$

D1 records a relatively high readiness score driven primarily by its strong current industrial density and superior logistics accessibility. The city has the highest industrial density and the strongest proximity to major logistics infrastructure, including roads, railway infrastructure, airport and railway station facilities, and port infrastructure. However, its expansion potential and temporal growth score are comparatively low. This profile reflects the characteristics of a mature industrial core: strong existing infrastructure and industrial concentration but limited remaining spatial capacity for major expansion.

D2 — Second Industrial City

$$IDRM(D2) = 0.20 \times (3.24 + 1.99 + 1.00 + 2.47 + 1.00) = 1.93$$

D2 obtains an intermediate overall readiness score. Its industrial density is considerably higher than D3, while its temporal growth is stronger than D1. However, its limited remaining undeveloped land and moderate logistics accessibility reduce its overall composite score. Its strong petrochemical and chemical specialization provides an important functional advantage and reinforces its role as an industrial processing hub.

D3 — Third Industrial City

$$IDRM(D3) = 0.20 \times (1.00 + 1.00 + 5.00 + 5.00 + 5.00) = 3.40$$

D3 achieves the highest overall IDRM score. Its high score is primarily associated with its substantial expansion potential, strongest absolute industrial growth, and strong industrial specialization. Although its industrial density and logistics accessibility are the lowest among the three cities, its spatial and temporal characteristics indicate a city with considerable future development potential. This result is consistent with Chapter 5,

where D3 was identified as a strategic industrial expansion zone characterized by emerging expansion and strong specialization in heavy industries.

Table 6.4. Final Industrial Development Readiness Scores

Rank	Industrial City	IDRM Score	Overall Interpretation
1	D3	3.40	Highest development readiness
2	D1	2.96	Strongest current industrial position
3	D2	1.93	Intermediate/transitional readiness

The results reveal an important distinction between current industrial strength and future development readiness.

D1 remains the strongest city from the perspective of current industrial concentration and logistics connectivity. D3, however, achieves the highest overall readiness score because the model gives equal consideration to expansion potential, temporal growth, and industrial specialization in addition to current spatial conditions.

This distinction is particularly important for planning interpretation. A city with a high current industrial density is not necessarily the city with the greatest capacity for future development. Conversely, a city with low current density may possess substantial undeveloped land and strong growth dynamics that increase its strategic development readiness.

CURRENT INDUSTRIAL STRENGTH ≠ FUTURE DEVELOPMENT READINESS

The IDRM should therefore be interpreted as a comparative decision-support framework rather than as a direct measure of absolute industrial performance.

6.7 Decision Support Dashboard

Power BI Decision-Support Interface. The analytical outputs generated throughout this study — spatial distribution, industrial growth, sectoral specialization, logistics accessibility, and the Industrial Development Readiness Model (IDRM) — were integrated into an interactive decision-support interface to enable comparative assessment of the three industrial cities. Rather than functioning as a data-visualization exercise, the interface was developed as the final decision-support layer of the IDRM framework, translating the preceding analytical findings into an explorable, comparative environment for planners and decision-makers.

The interface follows the same analytical flow established throughout this study:

Spatial Analysis → Growth → Industrial Specialization → Logistics Accessibility → IDRM

This structure ensures that the interface does not merely present descriptive statistics, but allows the user to move from understanding the current industrial footprint to evaluating growth patterns, specialization, accessibility, and ultimately development readiness.

6.7.1 Dashboard Structure

The interface consists of four integrated analytical views, each addressing a specific decision question.

1. Executive Overview. Provides a high-level assessment of the industrial system for decision-makers who require a rapid understanding of the overall situation — summarizing the total industrial factory footprint, the readiness ranking of the three cities, historical industrial growth, and a comparison between current industrial strength and future development readiness. This view answers: What is the overall industrial development position of the three cities, and where should attention be focused?

2. Spatial & Accessibility. Focuses on the geographic dimension of industrial development using an ArcGIS Maps for Power BI visual to display the industrial footprint by industry type, complemented by factory-density comparisons and a logistics-accessibility assessment covering road, railway, airport, port, and oil-and-gas connectivity. This view answers: How is industrial activity spatially distributed, and how does location and accessibility influence the development profile of each industrial city?

3. Growth & Industrial Structure. Examines how the industrial system has evolved over time (2000-2025) and how its sectoral composition differs between the three cities, using Location Quotient (LQ) values to identify differences in industrial concentration across sectors and cities. This view answers: How has industrial activity evolved over time, and what industrial sectors characterize each city?

4. City Profiles & IDRM. Translates the preceding analyses into a city-level development-readiness perspective, presenting the final IDRM scores and rankings together with current industrial strength, future development readiness, and the individual indicator profile for each city. This view answers: What is the relative development readiness of each industrial city, and what characteristics explain its position?

By consolidating these four views into a single environment, the interface allows decision-makers to move from where industrial activity is concentrated, to how it has evolved, to what characterizes it structurally, to how accessible it is — and ultimately to how ready each city is for further development, without interpreting each analysis in isolation.

6.7.2 Visual and Interaction Design

The interface follows a dark, executive-oriented visual identity based on a deep navy foundation with white information text and three consistent city-specific accent colors:

- D1 — First Industrial City: Navy #1B365D
- D2 — Second Industrial City: Turquoise #1E8A8F
- D3 — Third Industrial City: Olive #6E7A3C

The three colors are used consistently across comparative charts so that the identity of each industrial city remains visually stable throughout the interface. A persistent sidebar navigation system connects the four analytical views, and a clear visual hierarchy positions high-level indicators prominently while detailed tables provide supporting evidence for analytical users.

6.7.3 Analytical Integrity

The interface was constructed directly from the established semantic model of the IDRM analysis. No new analytical indicators or independent calculations were introduced at the visualization stage; existing project outputs were represented according to their appropriate analytical grain — factory records as counts rather than sums, pre-calculated density and readiness measures at their appropriate city-level grain, and Location Quotient values represented comparatively across cities and sectors. The interface therefore functions as a visual representation of the analytical model rather than a separate source of analysis.

6.7.4 Decision-Support Value

The main contribution of the interface is the integration of multiple analytical perspectives into one decision environment. Instead of requiring the decision-maker to interpret separate GIS maps, growth calculations, specialization results, accessibility assessments, and readiness scores independently, the interface establishes a continuous analytical narrative from where industrial activity is located, to how it has changed, to what sectors characterize it, to how accessible it is, and finally to how ready each city is for further development. It therefore represents the final decision-support layer of the IDRM framework rather than simply a data-visualization product.

6.7.5 Final Role within the Project

The overall project is therefore understood as a multi-stage analytical pipeline rather than a Power BI exercise:



↓ Power BI Decision-Support Interface

Within this framework, the interface represents the final layer through which the results are communicated and explored. Its value is therefore derived from the analytical framework behind it: it consolidates the evidence generated throughout the project and transforms it into a structured environment for comparing the three industrial cities and supporting strategic interpretation.

6.7.6 Dashboard Access

The complete interactive decision-support dashboard developed for this study is available through the following link:

Power BI Dashboard:

<https://app.powerbi.com/view?r=eyJrIjoiazGRiNzQzYzgtNGRiNS00MTQxLWExNWQxZGY5ZWY3YzMyZmU4IiwidCI6IjUxNGZhYTE5LTJjODQtNGNlZi04YWU5LTJiOWRiY2U5MzNjZCIsImMiOiJ9>

This dashboard presents the Executive Overview, Spatial & Accessibility analysis, Growth & Industrial Structure trends, and the final IDRM city profiles described throughout Sections 6.7.1–6.7.5. Selected visual samples are included in the appendix to support interpretation of the interface structure and its role within the overall analytical pipeline.

Chapter 6 Summary

This chapter developed and applied the Industrial Development Readiness Model (IDRM) using five spatial and temporal indicators derived from the GIS analyses presented in the previous chapters: Industrial Density Index (IDI), Logistics Accessibility Index (LAI), Industrial Expansion Potential Index (IEPI), Industrial Growth Index (IGI), and Industrial Specialization Index (ISI).

The indicators represent complementary dimensions of industrial development. IDI captures current industrial concentration, LAI represents accessibility to strategic logistics infrastructure, IEPI represents remaining spatial capacity for development, IGI captures industrial growth between 2000 and 2025, and ISI represents the relative strength and breadth of industrial specialization.

All indicators were standardized to a common 1-5 scale using min-max normalization and integrated using equal weights of 0.20. The resulting composite scores provide a comparative assessment of industrial development readiness across D1, D2, and D3.

The assessment highlights an important distinction between current industrial strength and future development readiness. D1 demonstrates the strongest current industrial position due to its high industrial density and strong logistics accessibility. D3, in contrast, demonstrates the highest overall development readiness because of its substantial expansion potential, strongest temporal growth, and strong industrial specialization. D2 occupies an intermediate position between the two.

The IDRM therefore provides a comparative GIS-based framework for integrating multiple dimensions of industrial development and translating spatial and temporal evidence into a structured decision-support assessment.

Chapter 7 — Conclusion

7.1 Key Findings

This study developed a GIS-based framework for evaluating the spatial characteristics and industrial development readiness of three industrial cities in Dammam, Saudi Arabia: the First Industrial City (D1), the Second Industrial City (D2), and the Third Industrial City (D3). The framework integrated macro-level industrial density and land characteristics with micro-level logistics accessibility, spatio-temporal development, Kernel Density Estimation (KDE), Standard Deviational Ellipse (SDE), and Location Quotient (LQ) analysis.

The first major finding is the substantial spatial variation among the three industrial cities. D1 represents the strongest current industrial position, with the highest industrial density and the most integrated logistics accessibility. Its factory density reached 71.20 factories/km², compared with 42.00 factories/km² in D2 and 4.69 factories/km² in D3. These differences demonstrate contrasting levels of current industrial concentration and development maturity.

The second major finding concerns logistics accessibility. D1 demonstrated the strongest overall spatial accessibility to the evaluated logistics infrastructure, including roads, railway infrastructure, airport and railway station facilities, port infrastructure, and oil and gas infrastructure. D2 maintained moderate accessibility, supported particularly by strong road connectivity, while D3 exhibited the greatest spatial separation from most of the evaluated infrastructure components. These findings highlight logistics connectivity as an important differentiating characteristic among the three industrial cities.

The third major finding concerns industrial growth. The temporal analysis covering 2000-2025 revealed distinct development trajectories. D1 increased from 30 to 40 sampled industrial facilities, D2 increased from 19 to 40, and D3 increased from 0 to 40. The strongest absolute increase occurred in D3, followed by D2 and D1. This progression

indicates a gradual transition from an established industrial core toward newer areas of industrial development.

The fourth major finding concerns industrial specialization. The LQ analysis revealed distinct industrial profiles. D1 exhibited a relatively diversified specialization structure, D2 demonstrated a stronger concentration in petrochemical and chemical industries, and D3 showed a pronounced concentration in metal, iron and steel, together with a high LQ for the Other Manufacturing category.

The final major finding is provided by the IDRM. The integrated assessment demonstrates that the city with the strongest current industrial position is not necessarily the city with the highest overall development readiness. D1 benefits from current industrial concentration and logistics connectivity, whereas D3 achieves the highest overall readiness because of its expansion potential, temporal growth, and industrial specialization.

7.2 Industrial Development Readiness Assessment

The final IDRM assessment distinguishes the three industrial cities according to their complementary development profiles.

D1 represents a mature industrial core. Its high industrial density and strong logistics accessibility provide significant current operational advantages. However, its limited remaining spatial capacity and relatively stable temporal growth constrain its potential for major future expansion.

D2 represents an intermediate industrial processing environment. It combines moderate industrial density, continuous growth, and a clear petrochemical and chemical specialization. Its relatively limited remaining spatial capacity and moderate logistics accessibility reduce its overall comparative readiness.

D3 represents the strongest future-oriented development profile. Although its current industrial density and logistics accessibility are lower than those of D1, its substantial undeveloped land, strongest absolute industrial growth, and strong sectoral specialization result in the highest overall IDRM score.

Accordingly, the IDRM ranking should not be interpreted as a simple ranking of existing industrial performance. Instead, it represents a comparative assessment that combines current conditions with spatial capacity, temporal development, logistics accessibility, and industrial structure.

7.3 Planning Implications

The findings provide differentiated planning implications for the three industrial cities.

For D1, planning priorities should focus on optimizing the existing industrial core rather than relying primarily on large-scale spatial expansion. Improvements in operational

efficiency, infrastructure management, internal connectivity, and industrial modernization may provide greater value than additional land expansion.

For D2, planning should emphasize strengthening its role as an industrial processing and manufacturing hub. Improving logistics connections and supporting linkages between petrochemical, chemical, and downstream manufacturing activities could reinforce its existing industrial specialization.

For D3, planning priorities should focus on managing future industrial expansion while addressing its relative logistics limitations. The substantial undeveloped land and strong temporal growth provide considerable spatial development potential, while improvements in logistics connectivity could strengthen the city's long-term industrial competitiveness.

These implications demonstrate the value of differentiating planning strategies according to the spatial and temporal characteristics of each industrial city rather than applying a uniform development strategy.

7.4 Research Contributions

This study provides several contributions to GIS-based industrial spatial analysis.

First, it developed an integrated GIS framework combining macro-level and micro-level analysis to examine industrial development from complementary spatial perspectives.

Second, the study integrated spatial, temporal, and sectoral analyses through industrial density, logistics accessibility, spatio-temporal analysis, KDE, SDE, and LQ.

Third, the study developed the Industrial Development Readiness Model (IDRM), which integrates five complementary indicators into a single comparative framework.

Fourth, the study demonstrates how GIS-derived evidence can be transformed into a structured decision-support product through the integration of spatial analysis, indicator standardization, comparative assessment, and interactive dashboard visualization.

Finally, the framework provides a transferable methodological structure that can be adapted to other industrial areas where comparable spatial, temporal, and industrial datasets are available.

7.5 Limitations

Several limitations should be considered when interpreting the findings.

First, the micro-level industrial facility dataset consisted of an author-compiled sample of 120 factories, with 40 sampled factories representing each industrial city. This sample was used for facility-level temporal, accessibility, and industrial specialization analyses and should not be interpreted as the official total number of factories within each industrial city.

Second, the logistics accessibility analysis used predefined 5 km, 10 km, and 15 km buffer thresholds. These thresholds provide a consistent comparative framework but may not fully represent actual travel time, transportation costs, road-network impedance, or freight accessibility.

Third, the IDRM applied equal weights of 0.20 to all five indicators. This provides a transparent baseline but does not imply that all dimensions have identical importance in every planning context.

Fourth, the IEPI represents remaining spatial capacity based on undeveloped land within the industrial-city boundaries. It does not independently account for land ownership, environmental restrictions, infrastructure capacity, economic feasibility, or regulatory constraints.

Finally, the IDRM is a comparative model for the three industrial cities included in this study. Its scores should therefore not be interpreted as absolute measures of industrial performance or as direct predictions of future development.

7.6 Practical Applications

The proposed IDRM framework may support industrial city authorities, infrastructure planners, regional development agencies, and investment promotion organizations in comparing industrial areas, identifying development priorities, and supporting evidence-based infrastructure investment decisions. The framework is intended as a comparative decision-support tool rather than a replacement for detailed feasibility assessment.

7.7 Future Research

Future research could extend the proposed framework in several directions.

First, expert-based weighting approaches such as the Analytic Hierarchy Process (AHP) or other Multi-Criteria Decision Analysis (MCDA) techniques could be applied to examine alternative weighting structures.

Second, future studies could incorporate formal sensitivity analysis to evaluate how changes in indicator weights and accessibility thresholds influence the final readiness ranking.

Third, additional spatial constraints could be incorporated into the expansion assessment, including environmental sensitivity, land-use compatibility, infrastructure capacity, and development regulations.

Fourth, the temporal component could be updated using post-2025 industrial data to monitor continuing changes in the industrial system.

Finally, future applications could integrate the GIS framework more deeply with interactive dashboard environments to enable scenario-based exploration and continuous monitoring of industrial development readiness.

7.8 Final Conclusion

The findings demonstrate that industrial development readiness cannot be adequately represented by a single measure of existing industrial intensity. The three industrial cities exhibit fundamentally different development profiles.

D1 represents the strongest current industrial position, supported by its high industrial density and strong logistics connectivity. D2 occupies an intermediate position characterized by continuous development and a strong petrochemical and chemical industrial role. D3 represents the highest overall development readiness under the proposed IDRM framework, driven primarily by its substantial expansion potential, strongest absolute industrial growth, and strong industrial specialization.

The distinction between current industrial strength and future development readiness is therefore a central outcome of this study. It demonstrates the value of integrating current industrial conditions, logistics accessibility, spatial capacity, temporal growth, and industrial specialization within a single comparative GIS-based decision-support framework.

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Appendix A — Dashboard Visual Samples

This appendix presents selected visual samples from the Power BI decision-support dashboard developed for the Industrial Development Readiness Model (IDRM). The visuals correspond to the analytical views described in Sections 6.7.1–6.7.6 and illustrate how the interface supports comparative assessment of industrial density, logistics accessibility, temporal growth, sectoral specialization, and development readiness across the three industrial cities.

Figure A.1 — Executive Overview Dashboard View

High-level summary of official factory counts, development-readiness ranking, sampled industrial growth, and the overall industrial position of D1, D2, and D3.

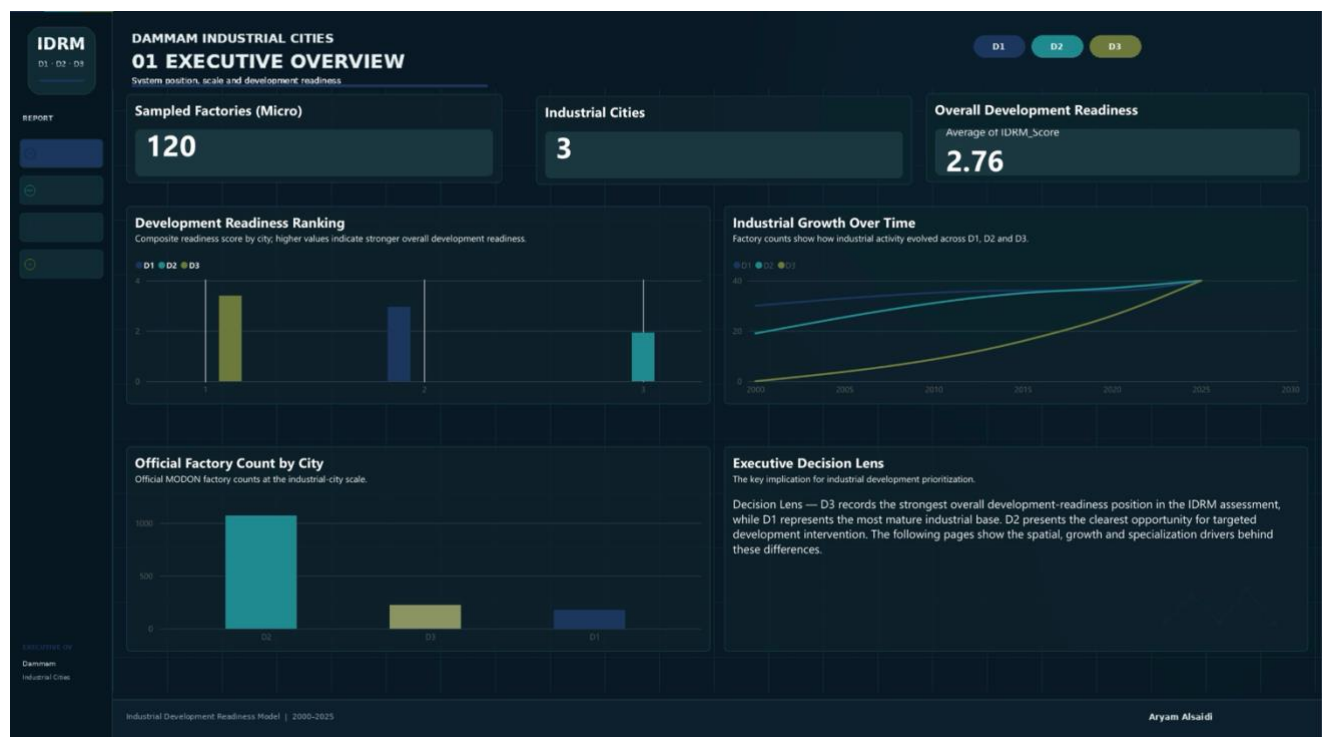


Figure A.2 — Spatial & Accessibility Dashboard View

Industrial footprint visualized using ArcGIS Maps for Power BI, including factory locations, sectoral distribution, density comparison, and accessibility to roads, rail, airport, port, and oil & gas infrastructure.

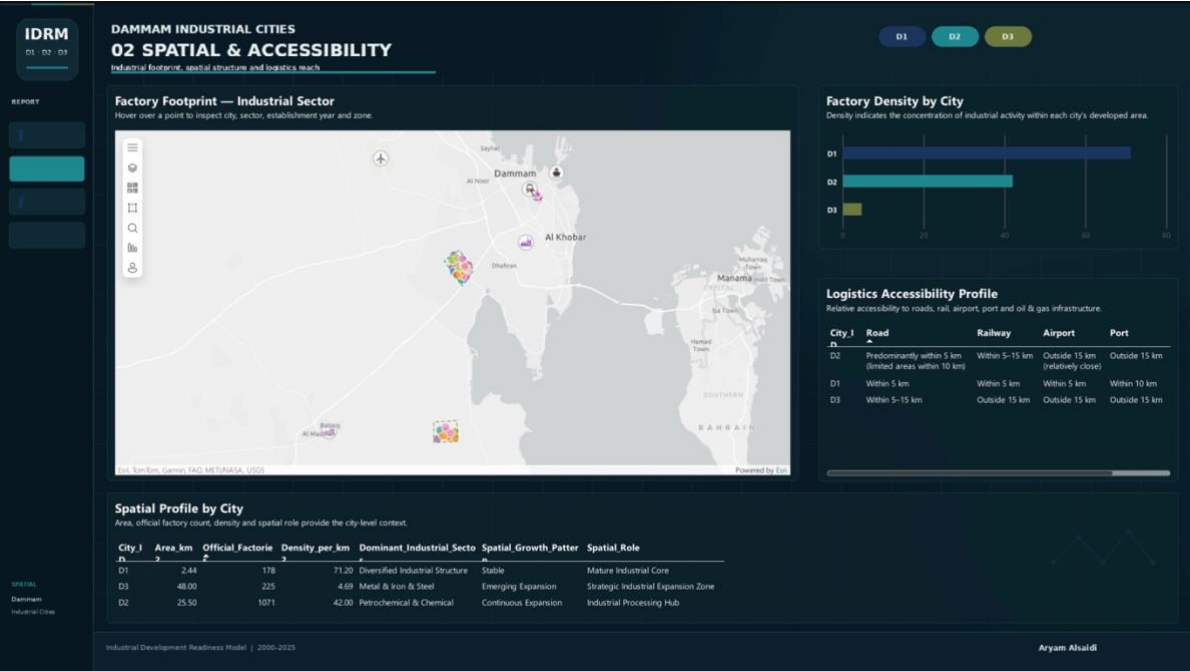


Figure A.3 — Growth & Industrial Structure Dashboard View

Temporal industrial growth (2000–2025), sectoral composition, and Location Quotient (LQ) specialization across the three industrial cities.



Figure A.4 — City Profiles & IDRM Dashboard View

Final IDRM scores, readiness ranking, current industrial strength, future development readiness, and standardized indicator profiles (IDI, LAI, IEPI, IGI, ISI).



Appendix Notes

- All dashboard visuals are extracted directly from the Power BI interface described in Chapter 6.
- The full interactive dashboard is accessible through the link provided in Section 6.7.6.
- Visual samples are included for documentation and interpretation purposes and represent the final decision-support layer of the IDRM framework.